Technological Forecasting & Social Change 182 (2022) 121838



Contents lists available at ScienceDirect

Technological Forecasting & Social Change

journal homepage: www.elsevier.com/locate/techfore





The impact of technology on sports – A prospective study

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ARTICLE INFO

Keywords:
Sportstech
SportsTech matrix
Technology
Sports
Delphi
Prospective study
Future study

ABSTRACT

Rapid technological progress and digitalization have considerably changed the role of technology in sports in the past two decades. As the human limits of performance have been reached in many disciplines, reaching future limits will increasingly depend on technology. While this represents progress in how athletes train and compete, similar developments await sports managers in the way they lead sports organizations and sports consumers in the way they consume and engage with sports. Using the SportsTech Matrix (i.e., a framework to capture how different types of technologies provide solutions to different user groups in sports), we examine how technology will impact sports in the future. We present a Delphi-based prospective study with quantitative and qualitative assessments from 92 subject matter experts for six future projections and 35 non-Delphi prospective survey items. We find that, by 2030, technology will significantly impact all three user groups in sports: athletes, consumers, and managers. For athletes, experts anticipate technology to play a major role for sporting performance improvements. For consumers, the consumption of sports content will continue to change significantly. For management, new types of manager profiles in terms of backgrounds and skill sets would be desirable. We discuss two possible future scenarios: (1) a probable future and (2) a game changer. Our findings should provide relevant insights for decision-makers and other stakeholders in sports and raise promising directions for future research.

1. Introduction

The role of technology in sports has drastically changed since the turn of the century in lockstep with technological advancements and digitalization across industries. Driven by the emergence of the Internet and mobile technologies, technology has become indispensable in many sports (Ratten, 2019). By playing "a vital role in becoming the best – on and off the pitch" (Schmidt, 2020, p. ix), technology has contributed to making sport potentially more exciting and challenging than ever before. While sports was considered a leisure time activity for most of its history, the technology-driven internationalization and professionalization have led to enormous changes: fans and consumers can follow sports events and their favourite teams and athletes across the globe live or on-demand through digital technology or social media basically at any time (Ratten, 2011). Athletes have more sports than ever to choose from and are much less dependent on other players or restrictions. In their quest for better performance, they have also benefitted from technology. Human limits seem to be reached in many sports (Berthelot et al., 2008, 2010; Nevill et al., 2007; Nevill and Whyte, 2005), and "future limits to athletic performance will be determined less and less by innate physiology of the athlete, and more and more by scientific and technological advances" (Lippi et al., 2008, p. 14). In most professional sports, it is assumed that any significant gains in the future will depend on technology to some degree (Balmer et al., 2012). Of course, such developments and prospects raise a variety of ethical questions that have been addressed in the literature as well (Dyer, 2015; Evans et al., 2017; Karkazis and Fishman, 2017; Loland, 2005; Miah, 2005).

Technologization in sports affects not only athletes and consumers but also sport managers who face an increasingly complex industry to manage and compete in. Historically, sports had been governed and managed through a variety of primarily public or voluntary organizations with a limited number of professional or commercial institutions. Given the rapid advancements and growth in the sports industry over the past decades, the pressure to adopt market mechanisms has constantly increased (Misener and Misener, 2017). Earlier non-profit organizations now have to be managed like profit organizations and establish professional organizational structures and processes (Kikulis et al., 1995; Skirstad and Chelladurai, 2011). In this complex development process, a variety of additional difficulties and limitations associated with non-profit organizations have to be addressed, such as resource insufficiencies, paternalism (i.e., decisions and resource allocation follow the interests of those responsible or most influential), or

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https://doi.org/10.1016/j.techfore.2022.121838
Received 15 August 2021; Received in revised form 1 May 2022; Accepted 20 June 2022
Available online 1 July 2022
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amateurism (i.e., managerial roles and tasks are taken over by well-meaning amateurs instead of qualified professionals) (Salamon, 1987). In addition, the globalization of sports poses challenges to sports managers ranging from a global flow of athletes to environmental impact and an increasing influence of international media corporates (Thibault, 2009). To manage this progress effectively, sports managers will continue to increasingly rely on technology (cf. Schmidt, 2020), and their work will need to be guided by innovation and entrepreneurship (González-Serrano et al., 2017).

In sum, almost all aspects of sports and all its stakeholders have been affected by technology in the recent past. Despite the high relevance in both social and economic terms, there are significant gaps in academic literature for sports technology (Ratten, 2017, 2018) as well as sports management (Gammelsæter, 2020; Peachey et al., 2015). Regarding the latter, there is a clear lack of research about sports managers; that is, while a lot of research and knowledge has been produced for sports managers, there is comparably little knowledge about sports managers, an issue that had been raised in the literature before (Andrews and Silk, 2018; Klikauer, 2018), and which is addressed in this research. Overall, the objective of this study is to understand better the possible role of technology in sports going forward, thereby contributing to the body of literature that currently discusses the expansion and integration of disparate research topics to provide more comprehensive perspectives (Ratten, 2020). In summary, the research question can be expressed quite simply: How will technology impact sports in the short, medium, and long term? Looking at this in detail, we are trying to answer several questions: Will the influence of technology decrease, remain constant, or increase? Will all players in the industry be affected in the same way? Will certain categories of technology play a more significant role than others? What opportunities are emerging and how can they be realized? What are the risks and how can they be mitigated?

Given this rather broad research question, we use the SportsTech Matrix developed by Frevel et al. (2020) to structure our thinking and research. The SportsTech Matrix provides an all-encompassing framework that captures the combination of technology and sports. For this purpose, all technologies relevant in sports are divided into three categories and all possible users of technologies in the sports context are divided into three groups. When put together, this allows to "capture how different types of technologies provide solutions to different user groups in the realm of sports" (Frevel et al., 2020, p. 25).

In this research, we conducted a real-time Delphi study among 92 subject matter experts from 30 different countries and heterogeneous backgrounds. Given the prospective nature of our study, Delphi represents an ideal research method as it allows examining a rather broad and complex issue both quantitatively and qualitatively. The latter is essential as it allows to tap relevant but tacit knowledge that is difficult to achieve in other research methods (Grisham, 2009). Delphi is a proven method in examining prospective developments of entire industries (Förster, 2015; Kluge et al., 2020; Merkel et al., 2016; Schuckmann et al., 2012; von Briel, 2018). To facilitate an effective group discussion among the experts (Belton et al., 2019; Linstone and Turoff, 2011), we decided to use the real-time variant of the Delphi method (Gnatzy et al., 2011; Gordon and Pease, 2006). To ensure a rigorous application of the research method with the highest standards, we have followed state-of-the-art methodological and technical papers from Beiderbeck et al. (2021b) and Schmalz et al. (2021), who suggest clear quality criteria for a three-step Delphi-procedure including study preparation, study conduction, and study analysis.

2. Background

To examine how the intersection of sports and technology may evolve and how technology will impact sports in the future, it is initially important to understand the general relevance of technology in sports and the relevance of technology for different user groups in sports, as these provide the basis for the formulation of our future-oriented projections.

2.1. General relevance of technology in sports

Depending on how it is defined, technology has always been of great importance in many sports. If technology is defined as "a design for instrumental action that reduces the uncertainty in the cause-effect relationships involved in achieving a desired outcome" (Rogers, 2003, p. 13), then technology is at the root of most sports as a requirement for existence given its constitutive function (Loland, 2009). Over time, technology has evolved from a constitutive function to a differentiating factor in performance improvements (Balmer et al., 2012; Foster et al., 2012; Haake, 2009; Stefani, 2012) as well as health and safety (Beiderbeck et al., 2020; Bjerklie, 1993; Miah, 2006; Schlegel and Hill, 2020; Waddington and Smith, 2000). Similarly, it has constantly gained relevance in the officiating of sports and adherence to the rules in many sports (Kolbinger and Lames, 2017) with inventions such as video-assisted refereeing (Tamir and Bar-Eli, 2021), the hawk eye in cricket or tennis (Collins and Evans, 2011), goal-line technology in football (Winand and Fergusson, 2018), or instant replay (Vannatta, 2011). The impact of technology on sports is no unidirectional relationship but rather a reciprocal one. That is, sports also have an impact on technology, typically by posing new challenges that are then being addressed with technology. For example, motorsports have produced a whole series of technological advances such as turbocharging, all-wheel drive, or traction control that are nowadays used in our everyday vehicles (Siegel and Morris, 2020). In machine learning and artificial intelligence, Chess has inhibited an important role as a natural testbed for innovations ever since Alan Turing created a chess computer program in 1948 (Hassabis, 2017; Heath et al., 1997).

Given this inevitable role of technology in sports, the above outlined lack of academic literature in sports technology (Ratten, 2017, 2018) is quite surprising. While this future-oriented research study at hand does not intend to provide a full-fledged review of the literature, it is important to note that the research field of sports technology could benefit from an extensive review that provides a structured overview of the different research streams in sports technology and the associated literature. To the best of our knowledge, there is no such literature review for sports technology overall available yet. However, there are several literature reviews that cover sub-domains of sports technology in different areas such as training (Rajšp and Fister, 2020; Simim et al., 2017), performance (van den Berg et al., 2020), rehabilitation (Cooper and Cooper, 2019), injury prevention (Tjønndal et al., 2022), health (Alqahtani et al., 2021; Duvall et al., 2021), motion capture (Jenny et al., 2017; van der Kruk and Reijne, 2018), biomechanics (Taborri et al., 2020), wearables (Aroganam et al., 2019; Mencarini et al., 2019), heart rate (Jiménez Morgan and Molina Mora, 2017), artificial intelligence (Bonidia et al., 2018; Ramkumar et al., 2021), machine learning (Cust et al., 2019), data collection (Owen et al., 2015), big data (Mataruna-Dos-Santos et al., 2020), virtual reality (Nor et al., 2020), additive manufacturing (Manoharan et al., 2013; Novak and Novak, 2020), innovation (Tjønndal, 2017), sport management (Tjønndal, 2016), sports fields (Straw et al., 2020), esports (Flegr and Schmidt, 2022; Reitman et al., 2019), sports betting (Winters and Derevensky, 2020), or ethics (Dyer, 2015).

In addition, we have performed a brief bibliometric analysis (Donthu et al., 2021) to construct and visualize networks of scientific publications (Fig. 1). We used Scopus to collect publications on the topic of sports technology (or related). The term sports technology (or synonyms) had to appear in the title, abstract, or keywords. This resulted in a total of 13.581 publications from the years 2002 to 2022. We then used VOSviewer (van Eck and Waltman, 2010) to build a co-occurrence map for author keywords. In total, 25.843 keywords were identified. Selecting a threshold of 30 occurrences left 100 keywords. The resulting map visually supports our notion, that the field of sports technology is still a combination of many subtopics rather than an established

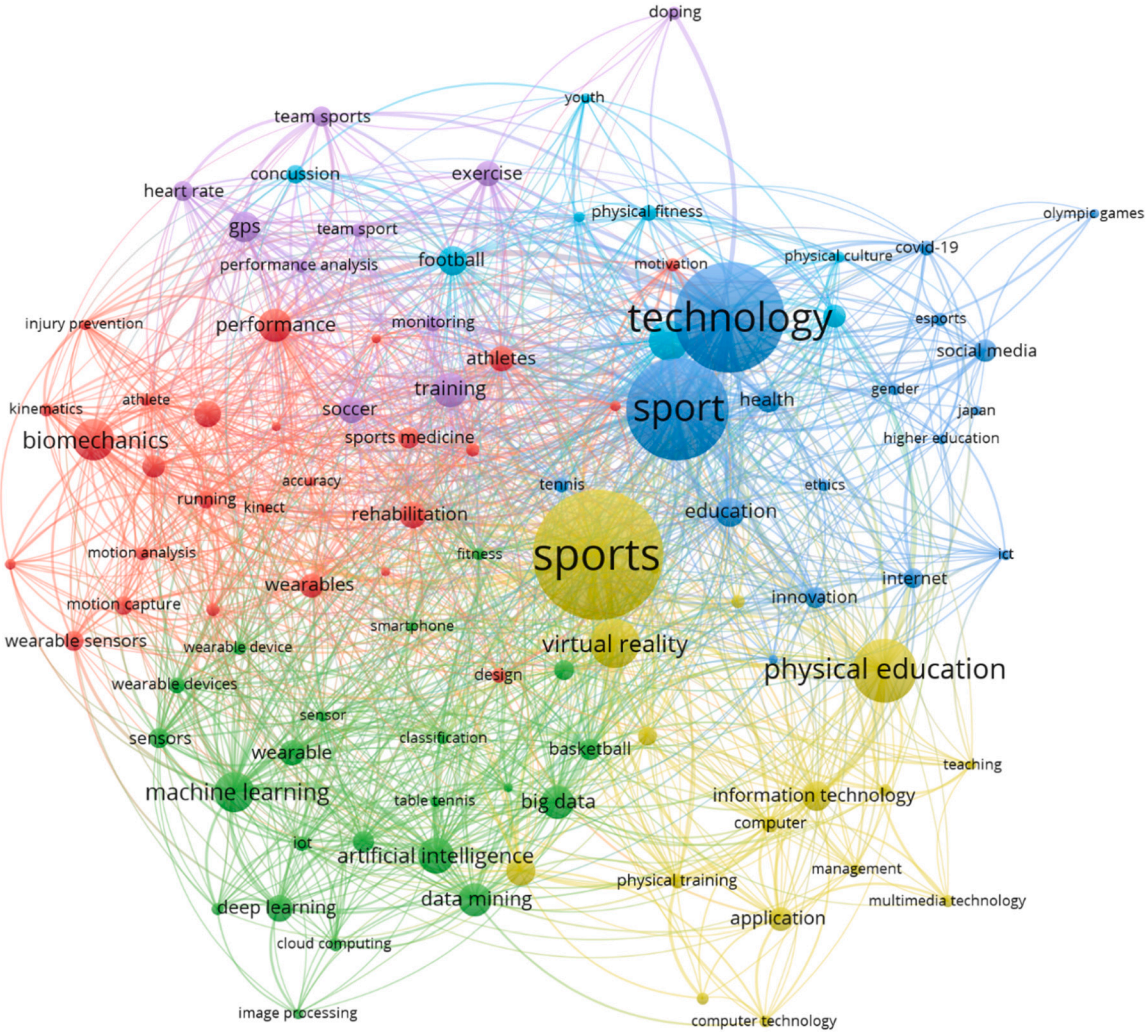


Fig. 1. Bibliographic co-occurrence map for author keywords.
Note. This co-occurrence map for author keywords was created using the bibliometric software VOSviewer (van Eck and Waltman, 2010). Data was collected from Scopus and is based on 13.581 publications between 2002 and 2022 with the term sports technology (or synonyms) in the title, abstract, or keywords. A threshold of 30 occurrences was selected, resulting in 100 keywords.

research stream in itself. The more frequently a keyword appeared in our Scopus data set, the bigger its circle. The lines represent links between keywords (i.e., if two terms occurred together in a publication, there is a link). In total, we found 1.409 links for the 100 keywords. Each keyword is assigned to a cluster following the VOSviewer algorithm, while each cluster (six in total) has its own color.

The overview of literature reviews and the bibliometric analysis support the need for two of our research objectives, namely, to gain valuable insights on the future developments in the field of sports technology and to facilitate an expert discussion in a yet unstructured field.

2.2. Relevance of technology for different user groups in sports

Following the structure provided by the SportsTech Matrix (Frevel et al., 2020), there are three different user groups to be considered with respect to sports technology: (i) athletes, (ii) consumers, and (iii) management. We can look at these user groups as being both exhaustive and mutually exclusive (i.e., any sports user belongs to one the three groups), where an individual can only represent one group at a time (naturally, an individual can be an athlete at one point and a consumer or manager at another). That is, with any given feature, a sportstech solution will always only address a user in one of the three user group roles. To shed more light on what this, we will elaborate on the relevance of technology for these user groups in the following.

For athletes, technology is not only at the root of many of their sports as seen above, but it is one of the most important means for them to improve their performance (Balmer et al., 2012; Berthelot et al., 2008, 2010; Crespo et al., 2022; Foster et al., 2012; Fuss et al., 2008; Haake, 2009; Lippi et al., 2008; Nevill et al., 2007; Stefani, 2012; Trabal, 2008; Whipp and Ward, 1992) and to maintain good health and safety (Beiderbeck et al., 2020; Bjerklie, 1993; Miah, 2006; Schlegel and Hill, 2020; Waddington and Smith, 2000). While health and safety was rather limited to physical factors for a long time, it increasingly includes mental health aspects (Chang et al., 2020; Ströhle, 2019; Stewart A. Vella et al., 2021; Stewart Anthony Vella and Swann, 2021). As suggested by Hudson et al. (2019), sports typically has the three phases of (i) pre-event with practice and training, (ii) performance (i.e., the actual competition itself), and (iii) post-event as the period after the competition which includes aspects such as regeneration as well as analysis which then ties back to phase (i) pre-event. Across all three phases, athletes may benefit from a wide variety of research that often stems from adjacent disciplines. This includes many forms of statistical analysis (Albert et al., 2017; Giblin et al., 2016; Lord et al., 2020; Perin et al., 2018; Sidhu, 2011), artificial intelligence in the analysis of sports performance (Araújo et al., 2021; Novatchkov and Baca, 2013), tactics

(Andrienko et al., 2021; Anzer and Bauer, 2021; Bauer and Anzer, 2021), practice-enhancing and human enhancement technologies (Farrow, 2013; Faure et al., 2020; Miah, 2006; Miles et al., 2012), sport-specific movement recognition (Cust et al., 2019; Minh Dang et al., 2020; Wu et al., 2021), biomechanics (de Magalhaes et al., 2015), human motion capture and tracking systems for sport applications (Adesida et al., 2019; Barris and Button, 2008; Taborri et al., 2020; van der Kruk and Reijne, 2018), integrated technology and microtechnology sensors (Cummins et al., 2013; Dellaserra et al., 2014; Rago et al., 2020; Whitehead et al., 2018), or ubiquitous computing in sports (Baca et al., 2009; Chi, 2008).

For consumers, technology enables all possible ways of engaging with a sport and it allows fans to interact on multiple levels (Schmidt, 2020). At the core is the provision of access to content, mostly through broadcasting of sport events, "which requires capturing, processing, and presenting the required information" (Frevel et al., 2020, p. 26). Technology provides consumers with data, analytics, and insights and it allows fans to engage with athletes, teams, brands, etc. by means of the internet, social media, etc. From a consumer perspective, initially, television was one of the most significant technological innovations for sports. With the maturation of television in the 1970s, an enormous dependency of sports on television (and vice versa) has emerged (Miah et al., 2020; Rowe, 1996; Whannel, 2014). Due to innovations such as mobile technologies, including smartphones and tablets, (mobile) Internet, or immersive technologies such as augmented reality and virtual reality, sports consumption has once again developed significantly in recent years and is expected to continue along this trajectory (Chan-Olmsted and Xiao, 2019; Hutchins, 2012, 2018; Miah et al., 2020; Pirker, 2020). The many ways consumers and fans may benefit from technology include enhanced traditional viewing experiences (Hutchins, 2018; Meese and Podkalicka, 2015; Wood and Begnini, 2006), access to more content through the internet and streaming providers (Ennis, 2020; Hutchins et al., 2019), augmented, virtual, mixed, or extended reality experiences (Kim and Ko, 2019; Kunz and Santomier, 2020; Zollmann et al., 2019), mobile devices (Chan-Olmsted and Xiao, 2019; Trivedi et al., 2021; Wang, 2021), ticketing (Khazaei Pool et al., 2016; Marquez et al., 2020), betting (Humphreys and Perez, 2020; Lopez-Gonzalez and Griffiths, 2018), and so on. Technology even enables access to new types of sports such as drone racing (Jung et al., 2018; Loquercio et al., 2020; Standaert, 2021) or esports (Carrillo Vera and Aguado Terrón, 2019; Hamari and Sjöblom, 2017; Pirker, 2020). While the amount of technology services in sports has risen substantially in recent years, further advances in technology will continue to change sports consumption significantly (Ratten, 2020).

For managers, technology plays an important role in helping them fulfilling their managerial or organizational roles and tasks in the realm of sports, including the management of associations, leagues, clubs, teams, events, venues, facilities, applications, plattforms, etc. (cf. Penkert, 2019). In addition, managers may also benefit from technology for "all aspects of sport governance that relate to defining rules and regulation or any other boundary condition in the context of sports" (Frevel et al., 2020, p. 27). As in any other industry, sports managers "at all levels will have to adapt to the world of smart machines" (Kolbjørnsrud et al., 2016, p. 2). The digitalization of jobs is one of the most important socio-technical transitions in our society (Govers and Amelsvoort, 2019) with a severe impact on sports managers, who will have to rethink the way they perform their work given this 'technochange' (technology-driven organizational change) (Markus, 2004). As a consequence, it is not only interesting to understand what technology-driven changes sports managers will need to manage, but also the effects of technology on their own roles and profiles (Duclos-Bastías et al., 2021). Going forward, innovation and entrepreneurship is likely to increasingly guide the work of sports managers (González-Serrano et al., 2017). The areas of technology application are then manifold and involve all relevant business areas, including human resources and people management (Chelladurai and Kerwin, 2018; Taylor et al., 2015), operations (Byers, 2004; Piekarz, 2020), finance (Horch and Schütte, 2003; Stewart, 2017), or strategy (Parent and O'Brien, 2018; Sarma et al., 2020; Schlembach et al., 2022). Technology can provide ample benefits to sports managers in their daily work in areas such as sponsorships (Dees, 2011; Koronios et al., 2020), events (Inversini et al., 2016; Parent and Ruetsch, 2020), ticketing (Courty, 2019; Popp et al., 2021), or scouting (Mataruna-Dos-Santos et al., 2020; Pavitt et al., 2021), to name but a few. This ongoing increase in relevance of technology in sports management has to a large degree been driven by the increasing professionalization and economization of the sports industry which fostered the embedding of technology in the sports industry (Ferreira et al., 2020; Miragaia et al., 2017; Ratten, 2020).

3. Research methodology

For this study, we used a mixed methods approach, including a Delphi study and 35 additional prospective survey items. For the Delphi study, we used a real-time format following approaches from Roßmann et al. (2018) and Beiderbeck et al. (2021b). To apply the Delphi method rigorously, we have been guided by numerous best practice papers in terms of methodical advice, including Beiderbeck et al. (2021a), Hasson and Keeney (2011), Markmann et al. (2020), and Schmalz et al. (2021). The 35 additional survey items were grouped into six questions and served to generate additional insights. The study can be divided into three major phases (Fig. 2): 1) study preparation, 2) study execution and expert selection, and 3) study analysis (Beiderbeck et al., 2021a). For data collection, we had a relatively long 13-week period between November 2020 and February 2021.

3.1. Study preparation

Since Delphi surveys are usually very time-consuming for participants, they are limited in scope and can therefore only include a limited number of projections, making thorough preparation all the more critical (Beiderbeck et al., 2021a; Schmalz et al., 2021). Clearly defined research objectives should be at the core of such preparation – with this study we have pursued three objectives: 1) Gain valuable insights on the future developments in the field of sportstech, 2) facilitate an expert discussion in a yet unstructured field, including a variety of stakeholders to create added value for everyone involved, and 3) add to the growing body of high-quality Delphi method literature by testing relatively new approaches of enriching Delphi studies. To achieve these objectives, we placed a high value on a careful composition of the (extended) research team, including non-authoring supporters in terms of subject expertise (sports industry and technology) and methodical expertise (Delphi method).

3.1.1. Initial conceptualization and creative workshops

The initial conceptualization was done by two members of the research team based on findings from initial desk research. The research scope was then narrowed done in a first of several creative workshop, which were mainly used to refine research scope, underlying theory or frameworks, and Delphi methodology specifics (Beiderbeck et al., 2021a). Concerning scope, we started very broadly with interest in understanding what the future of sportstech might look like in academia and practice. Global perspectives with experts from different geographies were included in the study preparation and in the study execution and analysis. If a Delphi study addresses a very broad field of interest, it is advisable to use a framework to guide the development of projections in a structured way (Beiderbeck et al., 2021b; Merkel et al., 2016). Therefore, we used the SportsTech Matrix developed by Frevel et al. (2020), which allowed us to systematically consider all user groups (user angle of the matrix) and the different types of technologies (tech angle) that are relevant in sports.

A real-time format of the Delphi method (Gnatzy et al., 2011; Gordon and Pease, 2006), was used, which is a variation of the traditional Delphi

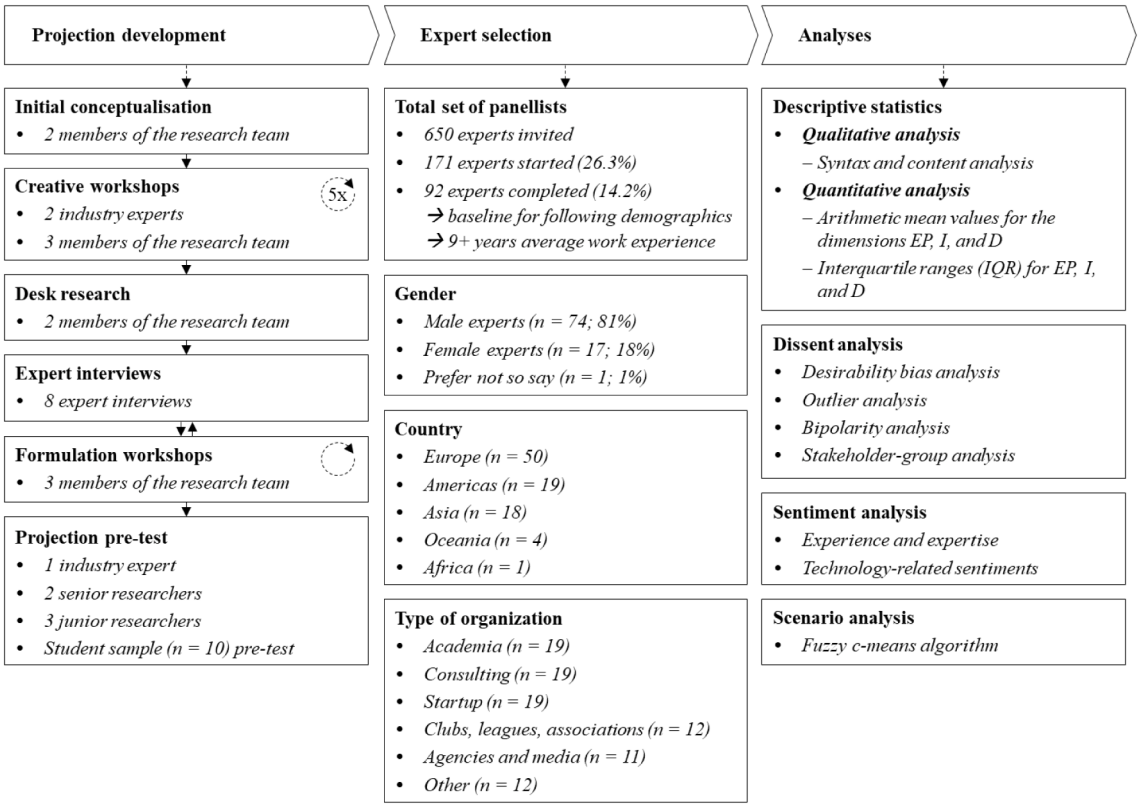


Fig. 2. Projection development, expert selection, and analyses based on Beiderbeck et al. (2021b) and Roßmann et al. (2018).

approach (Linstone and Turoff, 1975). It lets participants proceed with the questionnaire at their convenience with immediate access to other respondents' responses after completing the survey, thus permitting an effective online group communication process (Belton et al., 2019; Linstone and Turoff, 2011). Our main goal was to increase survey convenience, avoid research fatigue (Clark, 2008, 2010), and limit dropout rates between rounds. To execute the survey, we have chosen a proven software (Surveylet) as recommended in the literature (Aengenheyster et al., 2017).

3.1.2. Desk research and expert interviews

The development of the specific content of this study is primarily based on an iterative process of extensive desk research and numerous workshops and interviews with subject matter experts (Schmalz et al., 2021). First, we synthesized existing findings from academic literature supplemented with insights from industry reports, edited books and a series of relevant industry newsletters, as proposed by Beiderbeck et al. (2021a) for future studies where academic literature is scarce. Second, we tested, iterated, and supplemented the insights from our desk research with subject matter experts. Thus, we used expert input for both exploratory and confirmatory purposes (Bogner et al., 2014). To avoid biased perspectives, we ensured that the experts we engaged in interviews and workshops during the study preparation phase, had diverse backgrounds and represented different stakeholder groups, as detailed in Section 3.2. (Bonaccorsi et al., 2020).

In sum, this iterative process provided us with 205 factors or potential Delphi projections to consider. As suggested by Merkel et al. (2016), we used the chosen framework (SportsTech Matrix) to structure the factors and condense the long list of factors, thus allowing for an unbiased and balanced survey design (Nowack et al., 2011).

3.1.3. Formulation of Delphi projections

The formulation of the projections is one of the most crucial steps of a Delphi study and should be guided by state-of-the-art methodological research (Hasson and Keeney, 2011; Markmann et al., 2020; Schmalz et al., 2021). All projections in our study were formulated positively as recommended by Schmalz et al. (2021) and as concrete statements with simple and unambiguous wording in a single sentence (Markmann et al., 2020). To ensure and support this, we have placed great emphasis on sentence comprehensibility and concreteness. Keeping sentences short is essential as it promotes comprehensibility (Flesch, 1950), while longer sentences are typically more complex, making them more challenging to process and more time-consuming to absorb (Salancik et al., 1971; Stone and Parker, 2013). Longer sentences in Delphi studies often reduce the information value of the results as they were found to "lead to a dispersion around 'undecided'" Markmann et al. (2020, p. 11). In contrast, experts tend to be clearer in their assessments and responses towards shorter projections. In line with the generally recommended sentence length of 15–20 words (Cutts, 2020), we kept the core of all of our projections at <20 words.

For the time horizon of our study, we considered 2030 as not too far away to be unimaginable for participants but sufficiently far away to go beyond what might be only incremental developments. This ten-year time horizon goes beyond decisions that have already been made, thus stimulating outside-the-box thinking (von der Gracht and Darkow, 2010) and providing the possibility to take measures based on the findings. Similar Delphi studies have chosen similar time horizons (Merkel et al., 2016). To make sure participants consistently considered the right time horizon, each projection started with "By 2030, …" (Markmann et al., 2020).

The number of projections was intentionally limited to six as Delphi studies often tend to be overly time-consuming for respondents (Schmalz et al., 2021; e.g., Trevelyan and Robinson, 2015) and have to trade-off between gathering enough data and keeping it short (Schmalz et al., 2021). The less time-consuming and easier to answer a Delphi questionnaire is, the more likely respondents are to complete and revisit it (Hsu and Sandford, 2007). This challenge of panel attrition and participant dropout is one of the main threats to the validity of Delphi

results (Hasson and Keeney, 2011). It should be addressed with Delphi study designs that promote continuous participation of experts (cf. Bardecki, 1984). With our study, we are confident about having achieved this goal of developing a compact and engaging survey with both the Delphi projections as well as the additional prospective survey items.

In the final set of six projections, there were three projections each for the two respective angles of the SportsTech Matrix: User angle and tech angle. Together, both angles "provide an all-encompassing structure and a common understanding for the field of sportstech" by depicting the various types of technologies that offer solutions in sports and the different types of user groups that will use these solutions (Frevel et al., 2020, p. 24).

The user angle of the SportsTech Matrix was developed based on a framework (Malhotra, 2019; Penkert, 2017, 2019) that relied on a review of thousands of sportstech institutions to derive an exhaustive but mutually exclusive structure of user groups in sports: athletes, consumers, and management. Each user group is addressed with one projection. From our desk research and the expert input, we concluded that performance would be the most important topic for athletes, rather than focussing on any of the many potential sub-elements. Ultimately, all individual elements such as training and preparation, psychology, nutrition, or equipment advancements contribute to the athletes' performance which at the highest level is measured by world records (Lippi et al., 2008). For consumers, we aimed to focus on the most significant dimensions of all possible ways in which fans and consumers engage with a sport without performing (excluding the own performance of sports for this purpose). At the core of this idea is access to sports content which is why we developed a projection that allows for a fruitful discussion around how the consumption of sports content may change in the future. This could include not only a trend towards shorter sequences and more highlight-focused content (Hutchins et al., 2019) but also a development towards more personalized content using machine learning and AI (Chase, 2020) or data-enriched content, including live statistics (Van Gisbergen et al., 2020). With respect to sports managers, our expert discussions and desk research showed that one of the most interesting areas of investigation would not necessarily be the changes that sports managers need to manage as part of their jobs in the future, but how their own profiles might change (e.g., Duclos-Bastías et al., 2021). As a result, we derived the following three projections for the user angle:

Projection 1. By 2030, most of the current world records in Olympic disciplines will be broken due to technological advancements *(short: world records)*.

Projection 2. By 2030, consumption of sports content will continue to change significantly, for example, towards shorter sequences (highlights) and data-enriched content (live statistics, etc.) *(short: content formats)*.

Projection 3. By 2030, technological progress has led to new types of manager profiles in sports organizations (backgrounds, skill sets, etc.) *(short: manager profiles)*.The tech angle of the SportsTech Matrix was developed to categorize technologies in a way that is suitable to provide structure in the sports context (Frevel et al., 2020). Each of the resulting categories has a different focus, which we aimed to address with our projections. The first category is advanced materials, sensors, devices, Internet of things, and biotech and includes primarily physical technologies that are highly important in capturing data. Technology-based solutions for capturing data in sports have proliferated in recent years (Stein et al., 2017). Based on our desk research and the expert input, a focal question is whether the amount of data being collected is nearing a saturation point or whether it will continue to grow and, if so, how much. For the second technology category – data, artificial intelligence, and machine learning – our main area of interest was to understand better how automated data processing will develop over the next ten years. While the amount of data might be increasing, it is questionable whether it will be possible or feasible for athletes to use that data for relevant insights (Luczak et al., 2019). Advancements in automated data processing could be among the most considerable impact factors, for that matter. The third category is called information, communication, and extended reality and is mostly about human interaction. Our desk research and expert discussions have shown that AR/VR/XR and how frequently fans will use them to consume sports in ten years would be among the most interesting subjects for investigation (Miah et al., 2020). Overall, we concluded the following three projections for the tech angle:

Projection 4. By 2030, at least ten times as much data will be collected in professional sports compared to today *(short: data collection)*.

Projection 5. By 2030, automated data processing has significantly improved, allowing athletes to make better data-driven decisions around their sport (e.g., AI-generated training plans or optimization of technique) *(short: data processing)*.

Projection 6. By 2030, most consumers/fans will frequently follow (live) sports in AR/VR/XR *(short: consumers/fans)*.Before finalizing these six projections, we subjected them to rigorous testing by researchers, experts, and a student sample (Hasson and Keeney, 2011). First, in addition to prior expert input, we made a final check with one leading industry expert who is highly familiar with the global sportstech ecosystem. Second, we pre-tested the projections with two senior researchers and three junior researchers from our research network. Third, we had two Delphi expert members of our research team once again test all projections against all known quality criteria. Finally, we pre-tested the complete survey with a student sample from a sports strategy class of our MBA program. Twenty-five students were invited to participate, 18 started the survey, and ten fully completed it. This pre-test yielded good results and did not indicate any required changes.

3.1.4. Development of additional prospective survey items

Following the Delphi section of our survey, respondents had to reply to 35 additional prospective survey items grouped in six questions. In contrast to the Delphi projections, however, these questions had to be answered only once and were not asked again or displayed at all in subsequent revisits. This rather unique approach to Delphi-based prospective studies allowed us to address a very broad range of relevant topics without having to pose a full-fledged Delphi projection to experts for every single area of interest. For example, we were very eager to get the experts' assessment for individual technologies and individual user groups. We could do this with a matrix-like Likert scale based question instead of surveying 18 additional Delphi projections (three user groups times six technologies). This allowed us to keep the survey relatively short, as recommended by Schmalz et al. (2021), while at the same time avoiding the collection of unnecessarily large amounts of both quantitative and qualitative data, which can be a risk with Delphi projections. With this approach of relatively few Delphi projections combined with onetime prospective survey items, we aimed at avoiding research fatigue (Clark, 2008, 2010) and reducing dropout rates, which may threaten the validity of results (Hasson and Keeney, 2011).

The six additional questions that we posed focused on different topics identified as relevant following the same process that we used to develop the Delphi projections. Apart from the 18-item technology-centred question mentioned above, one 3-item question focused on the expected benefits from tech solutions for the three different user groups (athletes, consumers, managers). Another 3-item question assessed how much different players or institutions in sports would be impacted by technology in the next ten years. The next 3-item question addressed general developments of sports organizations, followed by a 4-item question about specific technology-enabled innovations or alterations. Finally, we asked respondents to rate their agreement on four items regarding possible developments by 2030 (e.g., "By 2030, investments (venture capital or private equity) into sportstech will have more than

tripled compared to today").

3.2. Study execution and expert selection

Given some respondents' unfamiliarity with the Delphi method, we carefully created the survey's landing page to ensure a proper survey process understanding among participants as recommended by Beiderbeck et al. (2021a). This included a brief explanation of the Delphi method, including a link to a short video and a brief explanation of common cognitive biases to sensitize participants to avoid them (Bonaccorsi et al., 2020). After the landing page, participants first had to respond to the six Delphi projections and then the 35 additional prospective survey items, which were again divided into six questions. After that, two sentiment constructs with a total of seven items were surveyed. Finally, participants were asked for demographics and three short self-assessments in terms of experience and expertise.

For the Delphi projections, respondents had to provide three quantitative assessments with expected probability of occurrence on an eleven-point scale from 0 to 100 % as well as desirability of occurrence and impact in case of occurrence on seven-point Likert scales anchored at "very low" (1) and "very high" (7). Like some other Delphi studies (Kluge et al., 2020), we chose seven-point Likert scales over more commonly used five-point scales given the seven-point scale's superiority in terms of reliability (Allen & Seaman, 2007; Nunnally, 1994). Following advice from Trevelyan and Robinson (2015), we required respondents to provide only one qualitative assessment at the end of each projection to provide reasoning for their assessment. We purposely limited ourselves to one qualitative input per projection to improve survey convenience and reduce research fatigue (Bardecki, 1984; Clark, 2008; Thiebes et al., 2018).

Delphi results' quality depends largely on the quality of experts (Förster and von der Gracht, 2014; Hasson et al., 2000; Nowack et al., 2011). A range of factors must be considered for expert selection, including the level of expertise, panel size, heterogeneity, and access to the panel (Beiderbeck et al., 2021a). Given our study's broad area of investigation, we aimed for a relatively large panel as suggested by Beiderbeck et al. (2021a). We ensured a heterogeneous composition in terms of backgrounds that help bring in diverse perspectives and improve the reliability of results and mitigate the risk of potential biases (Bonaccorsi et al., 2020). The diversity of backgrounds in our panel referred to different dimensions, including age, geography, job position, occupation, or stakeholder group (Mauksch et al., 2020). To identify experts, we followed recommendations from Kluge et al. (2020) and primarily relied on the research team's extended personal network of >30,000 contacts, including LinkedIn and direct contacts as well as desk research, including screening relevant interest groups or conference participation lists.

In total, 650 experts were identified and contacted individually via email or LinkedIn. 171 (26.3 %) started the survey and 92 (14.2 %) fully completed it, i.e., a response rate in line with similar Delphi studies (Beiderbeck et al., 2021b; Keller and von der Gracht, 2014). For an overview of panel composition, see Fig. 2. Experts were 39 years old on average and had slightly less than ten years of work experience. One expert preferred not to indicate their gender. From the remaining experts, 18 % were female, which seems to be a representative share given the low share of women in the sports industry (Burton, 2015) and the low share of women in the tech workforce which, for example, is below 20 % in 29 OECD countries included in the 2018 Women in Tech Index (Honeypot, 2018). Experts from 30 different countries and all five continents participated in the study, with 56 % from Europe, 20 % from Asia, 19 % from America, 4 % from Oceania, and 1 % from Africa. Occupational backgrounds were diverse and resulted in five larger groups: Academia (22 %), startup (22 %), consulting (21 %), adjacencies (20 %; including agencies, media, broadcasters, and others), and sports-centric (16 %; including sports associations, leagues, clubs, and athletes). All in all, we rate the panel as well-balanced and well-suited for the present study, especially given the typical reliance on convenience samples in such studies and the common array of difficulties in expert selection (Belton et al., 2019; Devaney and Henchion, 2018).

3.3. Study analysis

To derive multifaceted valuable insights from our study, we have performed a range of analyses (primarily using the open-source software *R*) that will be described in detail in the results and scenario section. The analyses can be categorized as descriptive statistics, dissent, sentiment, and scenarios. The descriptive statistics included qualitative analyses that contribute significantly to the interpretation of results such as a syntax and content analysis as well as quantitative analyses, including arithmetic mean value and interquartile range evaluations for the Delphi projections. While many Delphi studies in the past were focused on consensus, particularly prospective studies may benefit significantly from dissent analyses for valuable insights (von der Gracht, 2012; Warth et al., 2013). Consequently, we performed four different analyses for the dissent analysis: desirability bias analysis, outlier analysis, bipolarity analysis, and stakeholder group analysis. In addition to the dissent analysis, we evaluated experts' deep-level information to gain additional insights from our data, which had been recommended as early as in the 1980s (Loye, 1980) but was not commonly applied in Delphi studies (Spickermann et al., 2014). Given the context of our study, we include more common deep-level information such as experience and expertise, but also context-specific technology-related sentiments. Finally, we conducted a scenario analysis which is often a core feature of Delphi-based prospective studies (Nowack et al., 2011). All analyses will be described in more detail in the following.

4. Results

4.1. Descriptive statistics

In a first step, data was cleaned and checked for errors (Häder, 2009). A test for non-response bias following Roßmann et al. (2018) was performed to ensure the generalizability of results. To do so, we assumed late respondents and non-respondents to be similar (Armstrong and Overton, 1977) and applied Mann-Whitney *U* tests for comparisons between the first 20 (early) and last 20 (late) respondents across all dimensions of our six projections. There was no non-response bias present, based on the absence of significant differences between early and late respondents ($p < 0.05$).

For qualitative results, the respondents provided a total of 549 from 552 possible comments, i.e., an exceptional comment rate (99.5 %), indicating that our experts were very involved in the survey (Kluge et al., 2020). Syntax and content analyses (Table 1) were conducted to evaluate the quality of comments (Förster and von der Gracht, 2014). Like the overall comment rate, the share of whole sentences was particularly

Table 1
Syntax and content analysis of written statements.

Statement type	Total amount	Percentage share
Syntax analysis		
Whole sentences	512	93.3 %
Catchphrases	29	5.3 %
Catchwords	8	1.5 %
Blank	3	0.5 %
Content analysis		
Belief	216	39.1 %
Cause and Effect	152	27.5 %
Differentiation	102	18.5 %
Trend	57	10.3 %
Example	13	2.4 %
No information	12	2.2 %

high at 93.3 %, suggesting in-depth consideration and high involvement. From a content analysis perspective, beliefs (39.1 %) were most dominant, with cause and effect (27.5 %) and differentiation (18.5 %) as the next most frequent types of arguments and comments.

All six projections of our study achieved an impact rating between five ("somewhat high") and six ("high") on a seven-point scale, underlining the high relevance of the selected Delphi projections (Kluge et al., 2020). With an average of 5.85, projection 3 *(manager profiles)* yielded the highest impact in case of occurrence, whereas projection 1 *(world records)* had the lowest average impact at 5.21. Consensus on impact was reached for Projection 1 *(world records)* and Projection 2 *(content formats)*; the IQR threshold was 1.75 (25 % of seven-point Likert scale) (Warth et al., 2013). For an overview of the quantitative Delphi results, see Table 2.

Four out of six projections achieved a desirability rating between five ("somewhat high") and six ("high") on a seven-point scale and may thus be considered opportunities (Schweizer et al., 2020). Projection 1 *(world records)* – the least desirable projection with an average score of 4.77 and Projection 6 *(consumers/fans)* had a desirability rating of slightly below five, which is still "somewhat high" and therefore desirable from the experts' point of view. The highest desirability was assigned to Projection 5 *(data processing),* with an average desirability score of 5.75. Consensus was only reached for Projection 3 *(manager profiles)* at the same 1.75 IQR threshold as for the impact assessment. Finding consensus on desirability for only one of six projections is an interesting insight and may be interpreted as a selection of somewhat controversial projections. To address this in more detail and better understand potential underlying reasons, a rigorous dissent analysis was performed, as suggested by Warth et al. (2013).

All projections achieved an expected probability of occurrence (EP) of 60 % or higher, with Projection 4 *(data collection)* being the highest *(M = 88 %)* and Projection 1 *(world records)* being the lowest *(M = 60 %)*. Hence, the experts evaluated the projections as realistic future developments, which is a good result and underlines the relevance of the chosen projections. Consensus was reached for Projection 2 *(content formats)* and Projection 4 *(data collection)* at an IQR consensus threshold of 0.25 (i.e., 25 %) for EP estimates (Warth et al., 2013). Delphi projections with EP estimates at this level usually reach consensus (von der Gracht and Darkow, 2010). It is therefore all the more interesting that Projection 3 *(manager profiles)* and Projection 5 *(data processing)* have only almost reached consensus despite the high EP estimates of 79 % and 81 %, respectively. The lack of consensus for Projection 1 *(world records)* and Projection 6 *(consumers/fans)* with somewhat lower EP estimates is consistent with earlier Delphi research (Ogden et al., 2005; von der Gracht and Darkow, 2010); projections that are less likely to occur are typically more difficult to evaluate and often involve higher levels of dissent. These insights serve as a relevant basis for our dissent and sentiment analyses.

Upon the first completion of the Delphi projections, our respondents evaluated six additional prospective questions that included 35 Likert-type items. Comparing the results of early to late respondents, we found no non-response bias. The descriptive results for the additional prospective survey items can be found in Table 3 – they provide valuable input and insights for our analyses and discussions.

4.2. Dissent analysis

4.2.1. Desirability bias analysis

Desirability bias is a phenomenon that describes how the probability judgment of an event's occurrence may be influenced by the subjective opinion on the desirability (undesirability) of that event (Ecken et al., 2011; Krizan and Windschitl, 2007). The presence of desirability bias may skew probability judgments and consensus/dissent distributions, thus threatening the reliability of Delphi results and limiting their explanatory power (Ecken et al., 2011). To test for desirability bias, we followed the procedure suggested by Beiderbeck et al. (2021a) and evaluated Pearson's product-moment correlation coefficients between expected probability of occurrence and desirability for all six projections. The test showed correlations at the 5 % significance level for Projection 1 *(world records)* and the 1 % significance level for all other projections (see Table 4). As our study used a rather distant time horizon, the presence of a desirability bias is not necessarily surprising and in line with previous Delphi research (Beiderbeck et al., 2021b; Winkler and Moser, 2016). It is also worth mentioning that the potential existence of desirability bias in Delphi studies is mostly offset by Delphi's averaging process (Ecken et al., 2011). The bias is negligible if experts do not "share a pronounced and common desirability" (Ecken et al., 2011, p. 1666). Given that only one of our projections reached consensus on desirability and the average desirability scores of all projections were in a perfectly usual range, we concluded that the bias was negligible in our study. However, we still treat the results of this analysis as a relevant insight to keep in mind throughout our analyses and discussion.

4.2.2. Outlier analysis

To better understand the dissent in some of our projections and to test whether the mean values of our projections were biased or had inflated standard deviations due to outliers (Field, 2005), we conducted an outlier analysis by standardising the projections' EP values and testing whether they exceeded the threshold z-score of 2.58 (Warth et al., 2013). Only Projection 1 *(world records)* and Projection 6 *(consumers/fans)* had no outliers. For all other projections, we re-calculated EP mean values and the IQR after excluding the outliers from the data. The IQR remained unchanged in all four instances. The EP mean values, however, decreased by three percentage points for Projection 2 *(content formats),* two percentage points for Projection 3 *(manager profiles)* and Projection 4 *(data collection),* and one percentage point for Projection 5 *(data processing)*. Thus, while outliers may potentially explain some of the dissent in our results, the differences caused by these outliers are minor and can mostly be ignored, leading to the conclusion that no significant effect of outliers on our results is present. Our content analysis of the qualitative input further confirms this conclusion as no significant differences between outliers and other experts were found.

Table 2
Quantitative Delphi results.

Projection	EP				D				I			
	Mean	SD	Median	IQR	Mean	SD	Median	IQR	Mean	SD	Median	IQR
1: World records	0.60	0.24	0.60	0.40	4.77	1.39	5	2	5.21	1.17	5	**1**
2: Content formats	0.83	0.20	0.90	**0.20**	5.10	1.65	5	2	5.51	1.10	6	**1**
3: Manager profiles	0.79	0.22	0.80	0.30	5.70	1.24	6	**1.25**	5.85	1.09	6	2
4: Data collection	0.88	0.18	1.00	**0.20**	5.36	1.48	5.5	2	5.60	1.23	6	2
5: Data processing	0.81	0.20	0.85	0.30	5.75	1.24	6	2	5.84	1.14	6	2
6: Consumers/Fans	0.64	0.25	0.70	0.33	4.86	1.56	5	2	5.15	1.39	5	2

Note. EP = expected probability of occurrence (0–100 %); D = desirability of occurrence (7 pt. Likert scale; 7 = very high); I = impact in case of occurrence (7 pt. Likert scale; 7 = very high); SD = standard deviation; IQR = interquartile range; Consensus (IQR for EP $\leq$ 0.25, IQR for D and $I \leq 1.75$) is marked in bold.

Table 3
Quantitative prospective survey items results.

Question	Item	Mean	Median	Mode	SD
Please indicate how much the following technologies will impact athletes over the next 10 years. (1 = No impact at all, 5 = Full impact)	Advanced Materials	4.36	5	5	0.79
	IoT/Sensors	4.38	5	5	0.78
	AI/Machine Learning	4.26	4	5	0.83
	Advanced Analytics	4.41	5	5	0.89
	AR/VR/XR	3.18	3	2	1.16
	5G	3.29	3	3	1.16
Please indicate how much the following technologies will impact consumers/fans over the next 10 years. (1 = No impact at all, 5 = Full impact)	Advanced Materials	2.71	3	3	1.00
	IoT/Sensors	3.27	3	3	1.04
	AI/Machine Learning	3.63	4	4	1.18
	Advanced Analytics	3.66	4	4	1.07
	AR/VR/XR	4.17	5	5	1.04
	5G	4.43	5	5	0.94
Please indicate how much the following technologies will impact sports managers over the next 10 years. (1 = No impact at all, 5 = Full impact)	Advanced Materials	3.30	3	3	1.09
	IoT/Sensors	3.73	4	4	0.99
	AI/Machine Learning	4.27	5	5	0.91
	Advanced Analytics	4.64	5	5	0.69
	AR/VR/XR	3.15	3	4	1.13
	5G	3.63	4	4	1.16
Please indicate how much you think the following types of players/ institutions in sports will be impacted by technology in the next 10 years. (1 = Not at all impacted, 5 = Fully impacted)	Athletes	6.00	6	6	0.97
	Sports associations/ leagues/clubs	6.04	6	6	0.97
	Media and agencies (Sales, PR, etc.)	6.34	7	7	0.86
By 2030, the benefits from tech-solutions in sports will significantly increase compared to today for … (1 = Strongly disagree, 7 = Strongly agree)	... athletes	5.90	6	6	1.19
	... consumers/fans	6.13	6	7	0.87
	... sports managers	5.90	6	7	1.12
By 2030, most organizations in the world of sport will… (1 = Strongly disagree, 7 = Strongly agree)	... have a Chief Digital Officer (CDO) or Chief Technology Officer (CTO)	6.08	6	7	1.20
	... have hired/ employed new skills to manage technological advancements	6.34	7	7	0.94
	... be considered as leading-edge in terms of technology and innovation when compared to other industries	4.46	4.5	4	1.55
By 2030, new technology-enabled … (1 = Strongly disagree, 7 = Strongly agree)	... competition formats will have become normal (e.g., pro-level virtual indoor cycling races)	4.93	5	6	1.43
	... sports will have been established (cf. drone racing)	5.16	5	5	1.32
	... sports (e.g., drone racing or eSports) enjoy similar popularity as traditional sports	4.61	5	5	1.59
	... ways of fan engagement will have become normal (e.g., fans can virtually race against Formula 1 drivers)	5.51	6	6	1.26
By 2030, … (1 = Strongly disagree, 7 = Strongly agree)	... even more extreme versions of extreme sports will have emerged through technological advancements	5.09	5	5	1.28
	... sports tech will have significantly driven the development of new methods of training (e.g., cognitive training)	5.89	6	6	1.15
	... investments (venture capital or private equity) into sportstech will have more than tripled compared to today	5.97	6	7	1.13
	... new regulatory/ governance measures will have been put in place in many sports organizations to regulate the use of technological advancements in sports	5.86	6	7	1.26

Note. SD = standard deviation.

4.2.3. Bipolarity analysis

Another contributing factor to indications of dissent can be a bipolar distribution of data (Dajani et al., 1979). Thus, we evaluated whether the mean estimates for the projections' expected probability, desirability, and impact might stem from bipolar response distribution (i.e., one group providing high estimates and one group providing low estimates) (Warth et al., 2013). Following Linstone and Turoff (1975), we tested all dimensions of all projections for bimodal distribution and found two modes for the desirability assessments of Projection 2 *(content formats)*, Projection 4 *(data collection),* and Projection 6 *(consumers/ fans)*. However, in all three cases, both modes were very close together (maximum distance of two on a seven-point Likert scale), i.e., no indication of bipolarity. For expected probability and impact, there were no instances of bimodal distribution. A critical inspection of the histograms for all projections and all dimensions did not yield any visual cues that would suggest bipolar data distribution, leading us to conclude that bipolarity is not present in our data and does not help explain dissent in our projections.

4.2.4. Stakeholder-group analysis

Dissent in Delphi studies can be caused by diverging backgrounds from participating experts (Warth et al., 2013). To test for potential differences in assessments between different backgrounds, we conducted a stakeholder-group analysis for five distinct groups in our study: Academia ($n = 20$), startup (n = 20), consulting ($n = 19$), adjacencies (n

= 18), and sports-centric (n = 15); the results are presented in Table 5.

To identify significant differences between stakeholder groups in the non-normally distributed data, the nonparametric Mann-Whitney U test was applied (see Table 6). For groups *academia* and *startup*, we each found significant differences between their assessments of the projections and those of all other groups, respectively. Group *startup* had, on average across all six projections, the highest estimates for all three dimensions EP, D, and I as well as the highest EP and D estimates for five out of six projections and the highest I estimate for half of the projections (2: *content formats,* 4: *data collection,* 6: *consumers/fans*). Only for Projection 3 *(manager profiles)* were the EP and D estimates of group *startup* lower than that of another group. A potential explanation for this outcome could be that entrepreneurs often tend to be more optimistic and confident about the future (Koudstaal et al., 2015). Group *adjacencies* also provided relatively high estimates across almost all projections and dimensions, with the highest I estimate for Projection 1 *(world records)*, Projection 3 *(manager profiles)*, and Projection 5 *(data processing)*.

At the lower end of the spectrum, for half of the projections, group *sports-centric* had the lowest EP (1: *world records,* 3: *manager profiles,* 4: *data collection*) and D (1: *world records,* 2: *content formats*, 3: *manager profiles)* estimates; this group also had the lowest estimate for I of Projection 3. Group *adjacencies* also had several "lowest estimates," namely for EP of Projection 2 *(content formats)*, for D of Projection 4 *(data collection)* and Projection 6 *(consumers/fans*), and for I of Projection 2 *(content formats)* and Projection 6 *(consumers/fans*). Interestingly, group *adjacencies* had both a significant number of "highest estimates" and "lowest estimates," suggesting that this group felt somewhat stronger about the individual projections than other groups. A possible reason could be the diverse composition of the group *adjacencies* that is not necessarily united by a set of shared interests or disinterests. Group *academia* had the lowest estimates for EP of Projection 5 *(data processing)* and Projection 6 *(consumers/fans*), for D of Projection 5 *(data processing)*, and for I of Projection 4 *(data collection)*. Group *consulting* had the lowest I estimates for Projection 1 *(world records)*, Projection 3 *(manager profiles)*, and Projection 5 *(data processing)*.

Comparing individual groups bilaterally reveals additional interesting insights. For example, the largest difference in average EP estimates between any two groups was between group *startup* (EP = 0.73) and group *academia* (EP = 0.57) for Projection 6 *(consumers/fans*), which is not too far apart. Similarly, the largest difference in average D and I estimates is in both cases between group *startup* (D = 5.60; I = 5.55) and group *adjacencies* (D = 4.39; I = 4.83). These are interesting insights to bear in mind for subsequent analyses, results, and their discussion.

Since we found a substantial number of in some cases significant differences between groups, we did an in-depth examination of intra-group consensus for all stakeholder groups, all projections, and all dimensions (EP, D, I); see Table 7 for an overview. Intra-group consensus was reached in almost two-thirds of all possible instances. IQR scores were partially very low, indicating a very high level of consensus within groups in those cases. For example, group *sports-centric* had an IQR of 10 % for EP of Projection 2 *(content format)* and group *academia* had an IQR of 0.5 for D of Projection 3 *(manager profiles)*.

Table 7
Stakeholder group analysis – intra-group consensus across projections.

Stakeholder group	EP	D	I
Academia	1, 2, 3, 4, 5	1, 3, 4, 5, 6	1, 2, 3, 4
Startup	2, 3, 4, 5	1, 5	2
Consulting	1, 2, 4, 5, 6	3, 5	2, 3, 5, 6
Adjacencies	4, 6	1, 2, 3, 5	1, 2, 3, 4, 5
Sports-centric	2, 5, 6	1, 3, 4, 5, 6	1, 2, 3, 4, 5, 6

Note. Numbers indicate the projections for which intra-group consensus was reached; EP = expected probability of occurrence; D = desirability of occurrence; I = impact in case of occurrence. Numbers indicate respective projections.

4.3. Sentiment analysis

Following approaches from Beiderbeck et al. (2021b), Loye (1980), and Spickermann et al. (2014), we evaluated experts' deep-level information to gain additional insights from our data. In addition to the typical deep-level information variables of experience and expertise, we examined specific technology-related sentiments using established constructs for experts' attitude towards using technology (Teo and Zhou, 2014) and trust in technology (McKnight et al., 2011).

4.3.1. Experience and expertise

To assess our experts' experience and expertise, we relied on a multi-perspective approach (Nowack et al., 2011) that included years of work experience, job level, and stakeholder status as well as a self-assessment of expertise in closed-ended questions with the possible choices novice, beginner, intermediate, advanced, and expert. In addition, we ensured a certain level of expertise through the individual pre-selection of experts for our study (Förster, 2015). Finally, another valuable confirmation of our respondents' expertise was the critical review of comments (Bolger et al., 2011; Bolger & Wright, 2011; Bonaccio & Dalal, 2006). The sample may be considered quite experienced, with more than nine years of work experience on average. To check whether years of work experience influenced experts' assessments, we performed Mann-Whitney U tests for all relevant comparisons (see Table 8). We started by comparing only two groups with different levels of work experience in the sports industry: *moderate work experience* (less than five years; n = 33) and *significant work experience* (more than five years; n = 59). Experts with *significant work experience* group had rated the desirability of Projection 5 *(data processing)* and the impact of Projection 2 *(content formats)* and Projection 3 *(manager profiles)* significantly higher than experts with only *moderate work experience.*

To get a more nuanced understanding, we then further divided the experts and tested for differences in three distinct groups: *low work experience* (0 to 3 years; n = 15), *medium work experience* (3 to 10 years; n = 32), and *high work experience* (more than ten years; n = 45). A series of interesting findings came to light, for example, the more work

Table 5
Stakeholder group analysis.

Projection	Academia			Startup			Consulting			Adjacencies			Sports-centric		
	EP	D	I	EP	D	I	EP	D	I	EP	D	I	EP	D	I
1: World records	**0,62**	**4,45**	**5,15**	0,62	**5,35**	5,35	**0,62**	4,53	5,00	0,59	**5,17**	**5,39**	0,52	**4,27**	**5,13**
2: Content formats	**0,88**	4,85	**5,50**	**0,89**	5,80	**5,75**	**0,80**	4,84	**5,37**	0,76	**5,17**	**5,33**	**0,81**	4,73	**5,60**
3: Manager profiles	**0,82**	**5,90**	**5,95**	**0,78**	5,80	6,05	0,76	**5,47**	**5,53**	0,83	**6,11**	**6,11**	0,74	**5,07**	**5,53**
4: Data collection	**0,92**	**5,15**	**5,40**	**0,92**	6,00	5,85	**0,83**	5,37	5,68	**0,89**	5,06	**5,50**	0,79	**5,13**	**5,53**
5: Data processing	**0,76**	**5,50**	6,00	**0,88**	**6,10**	5,70	**0,82**	**5,74**	**5,68**	0,82	**5,67**	**6,00**	**0,78**	**5,73**	**5,80**
6: Consumers/Fans	0,57	**4,45**	5,25	0,73	5,60	5,55	**0,63**	5,00	**5,05**	**0,59**	4,39	4,83	**0,71**	**4,80**	**5,00**
Average	0,76	5,05	5,54	0,80	5,78	5,71	0,74	5,16	5,39	0,75	5,26	5,53	0,73	4,96	5,43

Note. EP = mean expected probability of occurrence (0–100 %); D = mean desirability of occurrence (7 pt. Likert scale; 7 = very high); I = mean impact in case of occurrence (7 pt. Likert scale; 7 = very high); Consensus (IQR for EP ≤ 0.25, IQR for D and I ≤ 1.75) is marked in bold.

Table 8
Sentiment analysis - Mann-Whitney *U* test results for all relevant comparisons for years of work experience.

Expertise area	Projection	Dimen-sion	Group 1			Group 2				
			Expert level	*n*	Mean	Expert level	*n*	Mean	*U*	*p*
Years of work experience	5: Data processing	D	>5 yrs	59	5.98	< 5 yrs	33	5.33	1217	0.038
	2: Content formats	I	>5 yrs	59	5.76	< 5 yrs	33	5.06	1322	0.003
	3: Manager profiles	I	>5 yrs	59	5.92	< 5 yrs	33	5.72	1168.5	0.097
	2: Content formats	EP	0–3 yrs	15	80 %	> 3 yrs	77	84 %	409.5	0.067
	2: Content formats	I	0–3 yrs	15	4.93	> 3 yrs	77	5.62	380.5	0.028
	3: Manager profiles	EP	0–3 yrs	15	71 %	> 3 yrs	77	80 %	419	0.089
	6: Consumers/Fans	I	0–3 yrs	15	5.73	> 3 yrs	77	5.04	751.5	0.060
	1: World records	I	3-10 yrs	32	4.88	< 3 or > 10 yrs	60	5.38	687.5	0.020
	2: Content formats	I	3-10 yrs	32	5.28	< 3 or > 10 yrs	60	5.63	757	0.079
	6: Consumers/Fans	EP	3-10 yrs	32	55 %	< 3 or > 10 yrs	60	69 %	660	0.013
	6: Consumers/Fans	D	3-10 yrs	32	4.34	< 3 or > 10 yrs	60	5.13	723.5	0.048
	2: Content formats	I	> 10 yrs	45	5.87	< 10 yrs	47	5.17	1457.5	0.000
	3: Manager profiles	I	> 10 yrs	45	5.98	< 10 yrs	47	5.72	1298	0.049
	5: Data processing	D	> 10 yrs	45	6.04	< 10 yrs	47	5.47	1267.5	0.087

Note. EP = expected probability of occurrence; D = desirability of occurrence; I = impact in case of occurrence.

experience experts had, the higher their impact estimate for Projection 2 *(content formats)*. In particular, experts with *high work experience* assigned a significantly higher ($M = 5.87$) impact to Projection 2 *(content formats)* than others ($M = 5.17$). One possible explanation could be that experts with less work experience tend to be younger and thus tend to be more digitally savvy, which may make new content formats more natural for them, leading to a low impact assessment. Another interesting finding is that experts with *medium work experience* assigned lower scores than other experts for the impact of Projection 1 *(world records)* and the expected probability and desirability of Projection 6 *(consumers/fans)*. That is, there was no tendency of correlation for these projections (e.g., impact assessment of the world records projection increases in work experience), but rather it is a u-shaped relationship. However, we did find three instances of significant positive correlation between work experience and our experts' assessments, namely for impact of Projection 2 *(content formats)* as well as expected probability and desirability of Projection 5 *(data processing)* (Table 9).

Job level may serve as another proxy for experience and expertise besides years of work experience. Looking at the four distinct groups of executives, senior management, middle management, and juniors, we checked for differences between these groups using Mann-Whitney *U* tests and found 13 instances of significant differences (Table 10). For example, executives estimated the desirability of Projection 1 *(world records)* and Projection 5 *(data processing)* as well as the impact of Projection 3 *(manager profiles)* and Projection 4 *(data collection)* significantly higher than non-executives. A possible explanation could be that they better understand certain industry dynamics or possess certain information that leads them to assign higher estimates to these projections. Conversely, juniors gave significantly lower estimates than others on the impact of Projection 2 *(content formats)* and Projection 3 *(manager profiles)*, which could be due to a lack of experience in evaluating such developments. They also had significantly lower estimates for the desirability of Projection 3 *(manager profiles)* and the expected probability of projection 5 *(data processing)*.

To look at more general results, we aggregated the existing four groups into only two groups ($N = 86$ as six experts indicated their job level as "Other" and were thus excluded from this analysis): *high job level* (executives and senior management; $n = 53$) and *low job level* (middle management and junior; $n = 33$). For all seven instances of significant differences that we found (Table 10), the estimates of experts with a *high job level* were higher than those of the experts with lower job levels. It is particularly interesting to see that they provided significantly higher desirability (projections 1, 3, and 4) and impact (projections 2, 3, and 4) ratings for half of the projections, respectively. This is an interesting finding which could indicate that experts with higher job levels are more optimistic, more experienced, or more informed than their lower-level colleagues. In line with these results, we have also found four

Table 9
Sentiment analysis – results of the correlation analysis for experience/expertise sentiments.

Parameter	Projection	Dimension	*r*	*t*	*p*
Years of work experience	2: Content formats	I	0.33	3.64	0.001
	5: Data processing	EP	0.18	1.70	0.092
	5: Data processing	D	0.25	2.42	0.018
Job level	1: World records	D	0.24	2.33	0.022
	2: Content formats	I	0.20	1.96	0.053
	4: Data collection	I	0.26	2.51	0.014
	5: Data processing	D	0.23	2.22	0.029
Technology experience/ expertise	1: World records	D	0.18	1.73	0.087
	2: Content formats	I	0.26	2.50	0.014
	4: Data collection	EP	0.24	2.39	0.019
	5: Data processing	EP	0.25	2.48	0.015
	5: Data processing	D	0.19	1.80	0.076
Sports industry experience/expertise	2: Content formats	I	0.24	2.35	0.021
	5: Data processing	D	0.19	1.91	0.060

Note. Statistics describe statistically significant correlations between sentiment parameters and EP, D, and I of each respective projection. EP = expected probability of occurrence; D = desirability of occurrence; I = impact in case of occurrence.

significant positive correlations between job level and expert assessments (Table 9), namely for the desirability of projections 1 *(world records)* and 5 *(data processing)*, as well as the impact of projections 2 *(content formats)* and 4 *(data collection)*. That is, the higher the experts' job level, the higher their estimates for these projections.

To test for expertise in the sports industry and the field of technology, we asked participants for self-assessments along five-point Likert scales, anchored at "novice" (1) and "expert" (5). For technology, more almost two thirds of experts had advanced ($n = 41$) or expert ($n = 18$) expertise, followed by intermediates ($n = 30$) and beginners (n = 3); there were no novices. Significant differences in the Mann-Whitney *U* tests were only found for groups expert and advanced in a total of seven instances (Table 11). Similar to our findings on job level, participants with expert

Table 10
Sentiment analysis - Mann-Whitney U test results for all relevant comparisons for job level.

Expertise area	Projection	Dimen-sion	Group 1			Group 2				
			Expert level	*n*	Mean	Expert level	*n*	Mean	*U*	*p*
Joblevel	1: World records	D	High	53	5.13	Low	33	4.33	1151	0.012
	2: Content formats	I	High	53	5.70	Low	33	5.09	1131	0.017
	3: Manager profiles	D	High	53	5.77	Low	33	5.45	1081	0.053
	3: Manager profiles	I	High	53	5.98	Low	33	5.61	1122	0.022
	4: Data collection	EP	High	53	89 %	Low	33	85 %	1080	0.045
	4: Data collection	I	High	53	5.87	Low	33	5.15	1140.5	0.012
	5: Data processing	D	High	53	6.04	Low	33	5.36	1153	0.010
	1: World records	D	Executive	35	5.17	Non-executive	51	4.59	1103.5	0.058
	3: Manager profiles	I	Executive	35	6.00	Non-executive	51	5.73	1078	0.089
	4: Data collection	I	Executive	35	5.91	Non-executive	51	5.37	1098.5	0.061
	5: Data processing	D	Executive	35	6.11	Non-executive	51	5.55	1106	0.050
	2: Content formats	D	Senior	18	4.22	Non-senior	68	5.26	396.5	0.012
	3: Manager profiles	EP	Senior	18	85 %	Non-senior	68	76 %	772	0.084
	4: Data collection	EP	Senior	18	96 %	Non-senior	68	86 %	824.5	0.013
	1: World records	D	Middle	18	4.22	Non-middle	68	4.99	449.5	0.078
	5: Data processing	D	Middle	18	5.17	Non-middle	68	5.94	411.5	0.026
	2: Content formats	I	Junior	15	4.87	Non-junior	71	5.59	307	0.007
	3: Manager profiles	D	Junior	15	5.27	Non-junior	71	5.73	387.5	0.082
	3: Manager profiles	I	Junior	15	5.47	Non-junior	71	5.92	307	0.007
	5: Data processing	EP	Junior	15	73 %	Non-junior	71	83 %	362	0.043

Note. EP = expected probability of occurrence; D = desirability of occurrence; I = impact in case of occurrence.

Table 11
Sentiment analysis - Mann-Whitney U test results for all relevant comparisons for technology and sports industry experience/expertise.

Expertise area	Projection	Dimen-sion	Group 1			Group 2				
			Expert level	*n*	Mean	Expert level	*n*	Mean	*U*	*p*
Technology	5: Data processing	EP	High expertise	59	83 %	Low expertise	33	78 %	1185.5	0.077
	5: Data processing	D	Advanced	41	5.51	Non-advanced	51	5.94	811	0.054
	1: World records	D	Expert	18	5.50	Non-expert	74	4.59	908.5	0.015
	2: Content formats	EP	Expert	18	88 %	Non-expert	74	82 %	854.5	0.056
	2: Content formats	I	Expert	18	6.17	Non-expert	74	5.35	943	0.004
	4: Data collection	EP	Expert	18	98 %	Non-expert	74	85 %	969.5	0.001
	5: Data processing	EP	Expert	18	92 %	Non-expert	74	79 %	922	0.010
	5: Data processing	D	Expert	18	6.44	Non-expert	74	5.58	982.5	0.001
Sports industry	2: Content formats	I	High expertise	71	5.66	Low expertise	21	5.00	999.5	0.013
	4: Data collection	EP	High expertise	71	89 %	Low expertise	21	81 %	939.5	0.049
	2: Content formats	I	Intermediate	18	5.06	Non-intermed.	74	5.62	488.5	0.066
	4: Data collection	EP	Advanced	29	92 %	Non-advanced	63	86 %	1124.5	0.053
	2: Content formats	I	Expert	42	5.71	Non-expert	50	5.34	1263	0.078

Note. EP = expected probability of occurrence; D = desirability of occurrence; I = impact in case of occurrence.

technology knowledge gave consistently higher estimates for the projections in all instances of significant differences. Particularly large were the differences for projection 5 *(data processing)*, where technology experts assessed the probability of occurrence ($M = 92$ %) and the desirability ($M = 6.44$) significantly higher than others (expected probability $M = 79$ %; desirability $M = 5.58$). Given the complexity of today's and future technologies, less tech-savvy experts might not yet grasp the full potential of technology or more tech-savvy experts might overestimate the application of technology in sports. Based on these insights, it is no surprise that the probability of occurrence and desirability of projection 5 *(data processing)* are significantly positively correlated with technology expertise and so is the probability of occurrence of projection 4 *(data collection)* (Table 9).

With respect to the sports industry, our participants had even more expertise, with almost half of them categorizing themselves as expert ($n = 42$), followed by advanced ($n = 29$), intermediate ($n = 18$), beginner (n = 2), and novice (n = 1). We found three instances of significant differences between these groups and aggregated participants into only two groups for a more conclusive comparison (Table 11): *high expertise* ($n = 71$) and *low expertise* ($n = 21$). Again, higher levels of expertise led to higher estimates and we found that participants with high sports industry expertise expected the probability of occurrence of projection 4 *(data collection)* significantly higher ($M = 89$ %) than participants with low expertise ($M = 81$ %). Likewise, they projected the impact of projection 2 *(content formats)* significantly higher ($M = 5.66$) than others ($M = 5.00$). Overall, the impact of projection 2 was positively correlated with sports industry expertise and the desirability of projection 5 *(data processing)* (Table 9).

4.3.2. Technology-related sentiments

For the examination of our experts' technology-related sentiments for additional insights (Beiderbeck et al., 2021a), we relied on two established constructs for attitude towards using technology (ATT) (Teo and Zhou, 2014) and trust in technology (TRUST) (McKnight et al., 2011) along seven-point Likert scales anchored at “strongly disagree” (1) and “strongly agree” (7). Both constructs were intentionally very short to limit assessment time and achieve high response rates as well as high face validity through lower perceived redundancy by respondents (Rammstedt and Beierlein, 2014; Wessel et al., 2019). A confirmatory factor analysis (CFA) was performed to confirm the constructs' measurement reliability and construct validity (Homburg and Giering, 1998). As threshold values served ≥0.6 for composite reliability (CR) and ≥ 0.5 for average variance extracted (AVE), according to the corresponding literature (Anderson and Gerbing, 1988; Bagozzi and Yi, 1988; Homburg and Giering, 1998). CR exceeded the 0.6 threshold proposed by Bagozzi and Yi (1988) for both constructs. Likewise, AVE

not only exceeded the pre-defined 0.5 threshold but even exceeded the preferred 0.7 threshold recommended by Nunnally (1994) for both constructs (Table 12).

First, to assess participants' attitude towards using technology (ATT), we adapted a construct from Teo and Zhou (2014) that builds on the technology acceptance model (TAM; Davis, 1989) and measures "the degree to which a user likes or dislikes using technology" (Teo and Zhou, 2014, p. 126). A reasonable hypothesis was to assume that experts with higher ATT scores provided higher estimates for the assessment dimensions EP, D, and I of our projections. When testing this hypothesis using Pearson's product-moment correlation coefficients between ATT for all dimensions of all projections, we found 13 (out of 18) statistically significant correlations (Table 13). For all six projections, desirability was positively correlated with ATT. Concerning expected probability, all except projection 6 *(consumers/fans)* were positively correlated. For impact, projections 2 *(content formats)* and 3 *(manager profiles)* were positively correlated. As all experts indicated fairly high levels of ATT, we could not assess differences between low ATT and high ATT groups but limited our analyses to correlations.

Table 13
Sentiment analysis – results of the correlation analysis for technology-related sentiments.

Parameter	Projection	Dimension	*r*	*t*	*p*
Attitude towards using technology (ATT); Davis, 1989; Teo and Zhou, 2014	1: World records	EP	0.28	2.82	0.006
	1: World records	D	0.35	3.49	0.001
	2: Content formats	EP	0.19	1.87	0.065
	2: Content formats	D	0.33	3.28	0.001
	2: Content formats	I	0.30	2.98	0.004
	3: Manager profiles	EP	0.28	2.73	0.008
	3: Manager profiles	D	0.26	2.60	0.011
	3: Manager profiles	I	0.22	2.16	0.033
	4: Data collection	EP	0.30	3.00	0.003
	4: Data collection	D	0.37	3.81	0.000
	5: Data processing	EP	0.26	2.59	0.011
	5: Data processing	D	0.28	2.80	0.006
	6: Consumers/ fans	D	0.22	2.13	0.036
Trust in technology (TRUST); McKnight et al., 2002; McKnight et al., 2011	1: World records	EP	0.25	2.50	0.014
	1: World records	D	0.44	4.60	0.000
	2: Content formats	D	0.21	2.08	0.040
	2: Content formats	I	0.35	3.60	0.001
	3: Manager profiles	D	0.22	2.13	0.036
	3: Manager profiles	I	0.36	3.65	0.000
	4: Data collection	EP	0.36	3.69	0.000
	4: Data collection	D	0.32	3.17	0.002
	4: Data collection	I	0.17	1.68	0.096
	5: Data processing	EP	0.30	3.02	0.003
	5: Data processing	D	0.32	3.25	0.002
	5: Data processing	I	0.28	2.73	0.008
	6: Consumers/ fans	D	0.20	1.95	0.054

Note. Statistics describe statistically significant correlations between sentiment parameters and EP, D, and I of each respective projection. EP = expected probability of occurrence; D = desirability of occurrence; I = impact in case of occurrence.

Second, to assess participants' trust in technology (TRUST), we adapted a construct from McKnight et al. (2011) that measures "the belief that the specific technology has the capability, functionality, or features to do for one what one needs to be done" and "the general tendency to be willing to depend on technology across a broad spectrum of situations" (McKnight et al., 2011, p. 7). Pearson's product-moment correlation coefficients between trust and all dimensions of all projections yielded statistically significant correlations for 12 out of 18 tests (Table 14). Interestingly, desirability was positively correlated with trust for all six projections; that is, the higher the trust, the higher the desirability. This is likely a consequence of the phenomenon that high-trust individuals typically expect better results from using technology (McKnight et al., 1998). Expected probability was positively correlated with trust for half of the projections (1, 4, and 5) and impact was positively correlated with trust for four out of six projections (2, 3, 4, and 5). To gain more nuanced insights, we performed Mann-Whitney *U* tests to compare *high-trust* ($n = 36$) and *low-trust* ($n = 25$) groups, where high trust was defined as a score larger than five and low trust was defined as a score smaller than four on a seven-point Likert scale. Consistently across all 12 instances of statistically significant differences between the two groups, the *high-trust* group gave higher estimates in their assessments, indicating trust in technology as a key factor in the belief in and openness towards technology. In particular, we found significant differences between the two groups in projections 1 *(world records)*, 4 *(data collection)*, and 5 *(data processing)* for expected probability, in projections 1, 4, 5, and 6 *(consumers/fans)* for desirability, and in all except projection 1 for impact. These findings could allow the conclusion that *low-trust* individuals would typically prefer less technological progress and perceive the impact of technology to be lower compared to their *high-trust* counterparts.

5. Scenarios and discussion

Scenarios are among the most useful tools for exploring possible and plausible future states of any given object of study. A scenario typically builds on several influencing factors to form an internally consistent and plausible narrative about possible future states (Van der Heijden, 2011; von der Gracht and Darkow, 2010). It is an anticipation of the future rather than a prediction. Scenarios are very useful in strategy-making, especially in environments influenced by external change (Van der Heijden, 2011). Scenario planning is rather new to the business environment (Bradfield et al., 2005) and to academia, with a substantial increase in publications after 2000 (Varum and Melo, 2010). Scenario analysis in prospective studies often builds on quantitative data for scenario formation and is then enriched by qualitative data. Therefore, scenario analysis is also very powerful in eliciting understandable insights from different types of research (Schmalz et al., 2021). For Delphi studies, this approach is well-established (Nowack et al., 2011) as it helps structure the discussion of results and the evaluation of qualitative data (Tapio et al., 2011).

To develop scenarios, we first identified the optimal number of scenarios by testing thirty validation indices based on the majority rule (Charrad et al., 2014); the result was two scenarios. We then tested two distinct cluster algorithms to detect structure in the data for average expected probability, desirability, and impact. Each resulting cluster would present one scenario. Fuzzy c-means clustering (FCM) (Roßmann et al., 2018) suggested the first scenario to consist of projections 1 *(world records)*, 2 *(content formats)*, and 6 *(consumers/fans)* and the second scenario to consist of projections 3 *(manager profiles)*, 4 *(data collection)*,

Table 14
Sentiment analysis – Mann-Whitney U test results for all relevant comparisons for trust in technology (TRUST).

Expertise area	Projection	Dimen-sion	Group 1			Group 2				
			Expert level	*n*	Mean	Expert level	*n*	Mean	*U*	*p*
TRUST	1: World records	EP	High	36	64 %	Low	25	49 %	295.5	0.023
	1: World records	D	High	36	5.53	Low	25	4.04	194.5	0.000
	2: Content formats	I	High	36	5.89	Low	25	4.88	259.5	0.003
	3: Manager profiles	I	High	36	6.42	Low	25	5.32	160	0.000
	4: Data collection	EP	High	36	93 %	Low	25	76 %	206.5	0.000
	4: Data collection	D	High	36	5.64	Low	25	4.60	284.5	0.013
	4: Data collection	I	High	36	5.72	Low	25	5.08	326.5	0.062
	5: Data processing	EP	High	36	88 %	Low	25	75 %	251	0.003
	5: Data processing	D	High	36	6.17	Low	25	5.40	272	0.006
	5: Data processing	I	High	36	6.11	Low	25	5.32	239.5	0.001
	6: Consumers/fans	D	High	36	5.19	Low	25	4.40	314.5	0.043
	6: Consumers/fans	I	High	36	5.39	Low	25	4.64	312	0.039

Note. EP = expected probability of occurrence; D = desirability of occurrence; I = impact in case of occurrence.

and 5 *(data processing)*. Similarly, hierarchical clustering using Euclidean distance and ward method (Kaufman & Rousseeuw, 2009; Murtagh & Legendre, 2014; Ward, 1963) suggested almost the same allocation of projections to scenarios. Only projection 2 was assigned to the other scenario (in FCM, the degree of membership of projection 2 to scenario 2 was also quite substantial at 30 %). After comparing the two resulting classifications, including a visual inspection, hierarchical clustering generated the most meaningful results (cf. Schmalz et al., 2021). Based on this analysis, we defined the first scenario, which consisted of four projections as "probable future" and the second scenario consisting of two projections as "game changer"; for an overview, including average assessments for expected probability, desirability, and impact on a scenario level see Table 15. An evaluation of these assessments combined with a visual inspection shows that the assignment of projections to scenarios is most strongly supported by expected probability, followed by desirability and impact, which is important to note for subsequent discussions. While both scenarios represent different potential futures, they are not mutually exclusive; thus, we also considered shared links and trends in our analysis of the individual scenarios.

5.1. Scenario one: probable future

Scenario one ("probable future") describes a highly probable future for sports with respect to technological developments and is composed of four projections *(2: content formats, 3: manager profiles, 4: data collection, 5: data processing)*. The experts consider these developments very probable, with projections having an expected probability of 79 % or above at an average of 83 %. They also consider them very desirable, with average desirability of this scenario's projections of 5.48 and a minimum of 5.10. Impact of this scenario was also assessed as very high at an average of 5.70 and a minimum of 5.51 (Table 16).

With the second-highest expected probability of all six projections (M = 83 %), experts reached consensus that the consumption of sports content will continue to change significantly until the year 2030. They also reach consensus that this will have a high impact (M = 5.51), however, they do not reach consensus for desirability, which is somewhat high on average (M = 5.10). These were the lowest impact and desirability assessments of all projections in scenario one. In their discussion, the experts put forth a whole range of reasons for change, including advanced technological opportunities, new media formats, and generational shifts. One of the biggest challenges in the next decade will be finding a way to cater to Generation Z consumers' demands (Yim et al., 2020). While they might not yet be the most financially attractive target group today, they could be in 2030, according to our experts. Sports media right holders will have to compete for consumer's attention and limited screen time with other (entertainment) formats – a development that we can already observe today (Lindholm, 2019). Social media has played a decisive role in this development and is expected to continue its influence. Many experts expect significant changes in the nature of content towards even more data richness, more immersive formats, and shorter sequences such as highlights. The expected increase in demand for exclusive formats, behind-the-scenes content, and similar features was mentioned less uniformly but clearly. An interesting finding in this context was the anticipated influence of video game standards for sports consumption in the future. It is worth mentioning that not all experts expect major changes, one group of experts expects these changes to come but only marginally. Many experts expect full-length games and live events to remain relevant, but their audiences might reduce. A big advantage of on-demand and more digital content is its availability for a truly global audience across time zones as well as other factors that might prevent fans from following sports, including outlier events such as the COVID pandemic.

Besides these more central discussion points, the experts raised several other interesting topics. By 2030, most professional athletes will be digital natives themselves, which might significantly add to how sports content is delivered. They also raised a series of interesting questions, such as at what level consumers' desire for more data and live statistics will be saturated, which leads to several further questions: Will consumers demand more data by 2030 compared to today? Will sports consumers demand more participation and impact on the game by 2030? Will all this data take some of the "art" out of sports? Moreover, if so, will this be a risk in terms of a negative selection bias early on in all youth academies?

The experts also acknowledged that not all of this is entirely new. A lot is already in motion but needs more time for diffusion and adoption. Many sports organizations are already producing their own digital offerings; however, they should be thinking even more like media companies. At the same time, some experts believe that the multitude of opportunities provides much room for disruptive innovations and the emergence of new multi-billion dollar companies in this market.

With respect to manager profiles, most experts agreed that technological progress will have led to new types of manager profiles in sports organizations in terms of backgrounds, skills, etc., by 2030. The panel did not reach consensus on the expected probability of this projection (M = 79 %), however, they did reach consensus on desirability, which had the second-highest assessment among all projections (M = 5.70).

Table 15
Scenario statistics.

Scenario name	Included projection (s)	EP (mean)	D (mean)	I (mean)
(1) Probable future	2, 3, 4, 5	83 %	5.48	5.70
(2) Game changer	1, 6	62 %	4.82	5.18

Note. EP = expected probability of occurrence; D = desirability of occurrence; I = impact in case of occurrence.

Table 16
Scenario 1 projections – main findings.

Projection	EP	D	I	Main findings
2: Content formats	**0.83**	5.10	**5.51**	Consumption of sports content will continue to change significantly towards even more data richness, more immersive formats, on demand, more digital, shorter sequences, exclusive formats, behind-the-scenes content, and similar features. Reasons include advanced technological opportunities, new media formats, and generational shifts. Full-length games and live events will remain relevant, but their audiences might reduce. Additional factors might add to these developments: increasing number of athletes will be digital natives, sports organisation will be thinking and operating more like media companies, video game standards might further influence developments.
3: Manager profiles	0.79	**5.70**	5.85	Technological progress will lead to new manager profiles in sports organizations. Experts considers this both desirable and impactful, given the lack of technology-related skills. The need for new skills towards more tech-savvy and analytical profiles is driven by new technologies, more data, competitive advantage, generation Z demands, investor demands, new business models, etc. Reskilling and upskilling will be important and there will be a lot more CIOs/CDOs/etc. compared to today. Potential hurdles could be unsuitable executive search criteria, uncompetitive compensation, and incumbent decision-makers. Therefore, it is not clear to what degree significant change will happen by 2030.
4: Data collection	**0.88**	5.36	5.60	By 2030, at least ten times as much data will be collected in professional sports. Main drivers include: easier and more affordable access to required technologies for professionals and amateurs, a higher level of granularity, more technologically advanced sports facilities and venues. Expert views diverge about the amount of data from slight increase to exponential increase. Competitive advantages are expected from the better use of data, making sense of data and extracting insights, not the sheer amount of data.
5: Data processing	0.81	5.75	5.84	Automated data processing will significantly improve allowing athletes to make better data-driven decisions. The most desirable and a high impact projection (experts view). Advantages/ Opportunities include: injury prevention, prolongation of careers, competitive advantages, better training, reduced dependence on coaches, access to coaching. Factors for diffusion include: type of sports, availability of (financial) resources, federations/leagues/clubs and their adaptation of such technologies, tech-savviness of coaches and athletes, etc. An important issue to be solved is the question of who should own the data.

Note. EP = mean expected probability of occurrence (0–100 %); D = mean desirability of occurrence (7 pt. Likert scale; 7 = very high); I = mean impact in case of occurrence (7 pt. Likert scale; 7 = very high); Consensus (IQR for EP $\leq$ 0.25, IQR for D and I $\leq$ 1.75) is marked in bold.

This was also the projection with the highest impact assessment (M = 5.85). The debate on this projection was less about whether technological advances and the associated requirements would change but whether this is expected to happen in sports organizations by 2030. The vast majority of experts acknowledged a lack of technology-related skills and a need for new skills towards more tech-savvy and analytical profiles. The drivers of this development include new technologies, more data, competitive advantage, generation Z demands, investor demands, new business models. Possible solutions include replacing managers with new profiles, expanding existing management teams with new profiles, and reskilling and upskilling existing managers. The experts have pointed out that some of these developments are already underway with many positive examples, however, the pace of change will increase over the next decade, according to them. For example, roles such as Chief Data Officer or Chief Innovation Officer are currently a rarity but could be the standard in 2030. On the other hand, traditional hiring frameworks could hinder this progress. This refers to the competencies and profiles that are being searched for and the compensation that is being offered; attracting and retaining employees with relevant backgrounds could be difficult if they are not compensated competitively. Another main hurdle that experts in the debate addressed was that the incumbent decision-makers might have little interest in giving up power to these new types of manager profiles. This is one reason why some experts do not expect such changes to take place on a large scale until 2030, but only later. These findings are a valuable contribution to existing literature as this provides knowledge about sports managers, not only for managers – a lack in research that had recently been criticized (Andrews and Silk, 2018; Klikauer, 2018).

With the highest expected probability (M = 88 %), the panel reached consensus that, by 2030, at least ten times as much data will be collected in professional sports compared to today. On average, the experts consider this both desirable (M = 5.36) and impactful (M = 5.40). The main drivers they see for more data include easier and more affordable access to required technologies for professionals and amateurs, more technologically advanced sports facilities and venues, and a higher level of granularity, creating significantly more data points (e.g., positional data of athletes during competition and training). About the actual amount of data, opinions differ. Some consider the current amount of data to be already very high and doubt a tenfold increase or see no reason for it, whereas others assume an exponential development of the amount of data. Accordingly, some experts also assume that a tenfold increase will occur well before the year 2030. What was more decisive from the experts' point of view is the relevance of the data. It is not simply a matter of generating more data. It must also have a clear benefit. Sports organizations, athletes, and other users of that data need to be able to make sense of the data and extract insights; this starts with asking the right questions to solve problems and identify opportunities. A data-related competitive advantage will most likely result from the ability to use the data rather than from the sheer amount of data. Therefore, the experts pointed out how important it will be to have athletes, managers, coaches, and staff who possess the relevant skills and train them. Data overload might become a problem for some stakeholders. The experts' discussion also raised some questions to which answers must be found: Who will own which data? Who will be allowed and able to monetize and how? How to ensure fair competition with different levels of access to technology and data? Do gathering and handling data require more regulation?

According to our panel, the most desirable (M = 5.75) projection was that automated data processing has significantly improved by 2030, allowing athletes to make better data-driven decisions around their sport. The experts also assigned this projection the second-highest impact (M = 5.84) and a high probability (M = 81 %). Mostly, the experts expect significant advantages and opportunities for athletes, for example, for injury prevention, prolongation of careers, competitive advantages, or better training overall. According to the experts, this could also significantly reduce athletes' dependence on their coaches,

which relates to training management and feedback and the quality and availability of coaches in general. At the same time, this will also place new demands on coaches in tech-savviness and data analytics. A key part of the discussion was which athletes would be affected at all. On the one hand, some experts argued that it would be more for professional athletes, as the application and necessary marketability would be difficult in lower leagues. On the other hand, other experts argue that these solutions could provide opportunities to amateur athletes who could not afford a human coach in the first place, i.e., the democratization of high-quality training, which the experts considered highly desirable.

There was a vivid debate on who should own the data and who should have access to it. Some experts argue that most data belong to the athletes, while others counter that significantly more insights could be derived if data was public and could be combined in larger datasets. Similarly, the question was whether that data could be used to generate accurate and valuable insights as there might still be a lack of coaches and athletes who can use it appropriately. The automation aspect in data processing could partly address this, however, many experts are still sceptical whether this level of automation will be possible by 2030. According to them, AI has not yet fully proven its potential for these purposes and they expect significant advancements to really take off only after 2030. They also pointed out that advancements will come much earlier in some sports than in others; factors may include available (financial) resources and how static versus dynamic the different sports are. The experts raised some other interesting topics for discussion. For example, the speed of progress depends largely on federations and leagues and their adaptation of such technologies and analytics. At the same time, data would have to be more athlete-focused and less for other stakeholders if it is to really add value for athletes. A key point that will play an important role in this development overall is that in 2030 almost all athletes will belong to a generation of digital natives and will have a much greater affinity for relevant technologies.

5.2. Scenario two: game changer

In contrast to the first scenario, scenario two ("game changer") describes technology-related developments in the future of sports that could be considered game changers to some extent. It is composed of two projections *(1: world records, 6: consumers/fans)* that are somewhat less probable, less desirable, and less impactful in the opinion of our experts compared to the projections from scenario one. However, this is not to say that they have a low probability, desirability, and impact. In fact, their expected probabilities of 60 % and 64 % are still relatively high compared to similar Delphi studies. The experts did not reach consensus on these two projections, which is reasonable given that projections with expected probability assessments of around 60 % tend not to reach consensus (Ogden et al., 2005; von der Gracht and Darkow, 2010). Impact and desirability were assessed by the experts to be somewhat high for both projections (Table 17).

With the lowest expected probability ($M = 60$ %), the lowest desirability ($M = 4.77$) and the second-lowest impact ($M = 5.21$), the panel did not appear to be fully convinced that most of the current world records in Olympic disciplines will be broken due to technological advancements by 2030. In the discussion, many topics were addressed, from human limits to regulation as well as positive and negative effects. One group of experts argued that human limits had been reached in many disciplines and there is not much that technology could help without significantly altering the nature of the sport or the like. Other experts counter that it is precisely for this reason that technology can now make the small but subtle difference it needs for marginal record improvements. However, this alone is not a particularly desirable state of affairs for the former group because they lose the desirable comparability of performances and records across generations. With respect to timing, some experts considered significant advancements possible until 2030 as the next generation of athletes are digital natives who could be much better in leveraging data and technology for their training and performance, while other experts do not expect any leaps in performance over the next years, but rather later. Some experts have pointed out the desirable benefits such as positive spill-over effects on amateurs or even society as a whole, for example, in terms of health. On the other hand, several possible negative consequences have also been addressed. For example, consumers and fans could lose interest if success in sports depended too much on technology and if the discrepancy between athletes would become too significant. This could be harmful to the Olympic brand in the eyes of some experts. Some experts expressed concern that too much technological advancement could lead athletes to exceed their limits in ways that could lead to injuries. These concerns, along with others, are part of many experts' advocacy for increased regulation through governing bodies. Depending on the outcome of such regulation (which will most likely differ between different sports), the impact of technology on sports and world records could be more or less significant. There appeared to be consensus between experts that the impact of technology may differ between different sports for various reasons. For example, some disciplines are already much closer to performance limits than others and some disciplines are, by definition,

Table 17
Scenario 2 projections – main findings.

Projection	EP	D	I	Main findings
1: World records	0.60	4.77	5.21	Experts are not fully convinced that most of the current world records in Olympic disciplines will be broken due to technological advancements by 2030. Pro arguments: technology will now make the subtle differences as human limits are reached, positive spill-over effects on amateurs and society (e.g., health), equipment in many disciplines (e.g., cycling, skiing, etc.) is still improving, etc. Counter arguments: further technological progress could alter the nature of sports and reduce the desirable comparability of performances and records, fans could lose interest, could be harmful to the Olympic brand, could lead athletes to exceed their limits (injury risk), etc. Experts consent that impact of technology on sports records will depend on regulation (federations, etc.) and differences between sports and disciplines (some are less technologically advanced than others).
6: Consumers/Fans	0.64	4.86	5.15	Consumers/fans will follow (live) sports in augmented reality (AR), virtual reality (VR), or extended reality (XR) by 2030, but not necessarily very frequently. The number of users of such technologies should not be overestimated. The time horizon might be too short. There is more optimism about AR than VR or XR. Larger breakthroughs will depend on significant improvements in the quality, availability, and affordability of technology. Hurdles: dizziness, lack of user-friendliness, existing VR solutions in sports have failed to meet expectations, dampened enthusiasm, high computer processing demands. Potential: ability to lift the sports experience to another level, taking the perspective of athletes, remote use cases, improve provision of data and information, new marketing options, new forms of competition, improving the participatory element for fans, innovating new sports, etc.

Note. EP = mean expected probability of occurrence (0–100 %); D = mean desirability of occurrence (7 pt. Likert scale; 7 = very high); I = mean impact in case of occurrence (7 pt. Likert scale; 7 = very high); Consensus (IQR for EP $\leq$ 0.25, IQR for D and $I \leq 1.75$) is marked in bold.

better suited to using technology to achieve performance advances. Overall, the experts assumed disciplines with more equipment (e.g., cycling) more likely to be impacted by technology than disciplines with less equipment (e.g., running).

The experts assessed the projection that most consumers/fans will frequently follow (live) sports in augmented reality (AR), virtual reality (VR), or extended reality (XR) by 2030, with the second-lowest expected probability ($M = 64$ %) and the second-lowest desirability ($M = 4.86$) of all projections. If it was to occur, experts believe this projection to have the lowest impact ($M = 5.15$), which was also reflected in their argumentation. For example, some experts argued that the number of users of such technologies should not be overestimated, especially with a time horizon of only ten years. This would also significantly depend on the diffusion of the required technology among consumers, primarily dependent on other (entertainment) industries. For this debate, experts differentiated between AR, VR, and XR and they were significantly more optimistic about the diffusion and adaptation of AR advancements than VR and XR. A major reason is that AR applications are often smartphone-based, which is already much more widespread than VR devices. Particularly with regards to VR, some experts argued that breakthroughs would depend on significant improvements in the quality, availability, and affordability of technology. Many people still experience negative effects such as dizziness or struggle with the user-friendliness of available devices and applications. In addition, some experts have cited existing VR solutions in sports that have failed to meet expectations, ultimately dampening enthusiasm for such solutions and slowing development. High demands in terms of computer processing capacity were named as an additional hurdle for faster adaptation.

The discussion also involved many advantages of such technologies and applications. Many experts agreed that these technologies had the potential to lift the sports experience to another level. For example, an unlimited number of fans, no matter where they are located, could get into the pleasure of first row experiences that would otherwise only be available to a very small group of physically present and mostly privileged people (Chase, 2020). It could allow fans to almost "live the experience," as one of the experts put it, including taking the perspective of athletes. The current COVID pandemic has made the potential of such remote use cases even clearer. Such technologies can significantly improve the provision of data and information. From a commercial perspective, this may bring new marketing options as well as disruptive applications for ticketing. Further advantages and opportunities mentioned by experts in the discussion include the possibility of competing against (professional) athletes virtually, improving the participatory element for fans, or innovating new sports and competition formats.

Finally, this debate has also raised a couple of questions that could not be answered conclusively. For example, will these technologies ever be able to fully replace the experience of physically going to the stadium? Will it be a generational question of who uses such technology and applications and who might not? Will the next generation of sports consumers demand such offerings, or will they be more traditional than expected in their consumption of sports content? One explanation why the Delphi panel was not able to find conclusive answers to these questions refers to the subjective perception of a potential "technology overdose". While innovations will presumably offer the technical possibilities to fully replace a stadium experience, for instance, it is questionable, if sports spectators will develop a habit to consume all their favourite sports digitally. At some point, the desire for purer forms of sport with less involvement of technology might become a very relevant factor (Hutchins, 2016), comparable to the phenomenon of deliberate reduction of the use of technology, which is typically referred to as "digital detox". While this phenomenon appears to be more present among generations that grew up with significant less technology, we perceive that also younger generations – digital natives – increasingly feel the need for a reduction of technology at times and we expect this to be even more relevant in the future.

6. Conclusion

6.1. Limitations and future research

To address the limitations of this study, we started by critically reviewing the composition of our expert panel as recommended by Ecken et al. (2011) and Kluge et al. (2020). This is important to avoid potential biases such as social desirability bias and self-interests of certain experts that could influence study results. From a geographical perspective, it would have been desirable to have an even more balanced panel. We saw an over-representation of experts from Europe who made up 54 % of all participants, followed by America and Asia with 20 % each. Oceania (4 %) and Africa (1 %) were under-represented. A likely explanation for this geographic distribution could lie in the composition of the authors' networks (including LinkedIn contacts) that were a key source in selecting participants. It is positive that experts from 30 different countries participated in the study. In sum, while the geographic distribution of participants could have been better, we still believe our results are a legitimate representation of the potential development of technology's impact on sports globally. Looking at their backgrounds, our experts were evenly distributed among the relevant stakeholder groups (academia 22 %, consulting 21 %, sports-centric 16 %, startup 22 %, and adjacencies 20 %). Nonetheless, individual groups might still have certain self-interests that could bias results by influencing the experts' assessments. Such self-interests or biases could include an overly optimistic reflection of the conditions and progress in one's own industry or one's own business interests that may lead to distorted assessments (cf. Kluge et al., 2020).

Another limitation of our study is the limited number of Delphi projections. Compared to many other Delphi studies, the total number of projections was relatively low. While the study results suggest that relevant and appropriate projections were selected, building on a broader set of projections would have presumably led to additional insights. For example, this could have provided more space to consider more extreme projections that would potentially have a low probability but high impact assessments and are thus important for stakeholders to have in mind. The inherent trade-off between study completion time and breadth of content is one of the most difficult decisions in Delphi research and we encourage future research to further look for ways to best achieve both goals.

This study and the SportsTech Matrix, in general, provide plenty of starting points for future research to investigate the impact of technology on sports. While we have taken a high-level overview in this study, future research could take more focused approaches. For example, it could be interesting not to consider all user groups in one study but focus on the impact of technology on sports separately for athletes, consumers, and management. Alternatively, researchers could approach the topic from a technology perspective, examining what role a particular category of technologies will play for user groups. Ultimately, the research could ideally become granular enough to examine in detail the effect of very specific technologies on very specific user groups. In order to conduct such specific research, using the Delphi method will depend enormously on the selection of experts, which could become increasingly difficult as the granularity increases.

It would be interesting to consider a wider range of time horizons in future research. Replicating our study or conducting similar studies with different time horizons could yield valuable insights. Shorter time horizons of five years or less could allow focusing on the most pressing issues, while much longer time horizons of 15–20 years or beyond could help to identify major threats and opportunities early.

Finally, this research could be used to derive several highly relevant and very concrete research questions for future work. For example, how do governing bodies want to approach the diminishing comparability of world records? Should there be more regulation to limit the influence of technology? Should one accept that some of the future world records will no longer be comparable to earlier world records? Or could there be

a dual system that acknowledges technology-enabled world records versus non-technology-enabled world records? It would also be interesting to look at the concrete required skills of sports managers in the future in terms of what will be really important and what could be nice to have; this could inform the training and education of sports managers. Research could also look at the trend of sports consumption shifting towards highlights instead of watching full competitions. Does this carry the risk that at some point it might no longer be economically interesting for broadcasters and media companies to produce full-length sports content? This could be an undesirable negative externality of market dynamics and an interesting area for investigation.

6.2. Summary

This study provides multifaceted findings on the impact of technology on sports in the future, which reflects a valuable contribution to the field given the profound impact of technology on sports (Schmidt, 2020), the large economic and social relevance of sports (Katsarova and Halleux, 2019; Lefever, 2012), and the lack of research on the future of sports (cf. Merkel et al., 2016). We presented the results of a Delphi-based prospective study with 92 subject matter experts. They gave their quantitative and qualitative assessment of six future projections in terms of expected probability, desirability and impact and 35 additional prospective survey items grouped in six questions, which only had to be answered once. This allowed us to address a broader range of relevant topics within a reasonable survey scope (Schmalz et al., 2021), overcoming a typical drawback of Delphi studies. We also gathered participants' deep level information, including experience and expertise as well as context-specific technology-related sentiments (Loye, 1980; Spickermann et al., 2014).

With the chosen Delphi design, we attempted to contribute to the continued development of the Delphi method and its rigorous application. By keeping it short and engaging (Schmalz et al., 2021; Trevelyan and Robinson, 2015), we addressed one of the most important threats to the validity of Delphi results, participant dropout and panel attrition (Bardecki, 1984; Hasson and Keeney, 2011). The amendment of non-Delphi survey items ensured the gathering of sufficient amounts of data (Schmalz et al., 2021). The important factor of an expert's personality on his or her assessment of future developments was taken into account through experts' deep-level information and sentiment analyses (Loye, 1980; Spickermann et al., 2014). Overall, we thus further developed the Delphi method in more specific formats that can be helpful in future research endeavors across research fields and we responded to Misener and Misener's (2017, p. 130) call for "more 'risky' research endeavors […] to embrace the complexity of contemporary sport management." At the same time, the rigorous methodological approach and presentation should have contributed to scenario planning and futures and foresight research (cf. Chermack, 2018; Fergnani and Chermack, 2021; Gary and Von Der Gracht, 2015; Münch and von der Gracht, 2021; Varum and Melo, 2010), which will hopefully help to establish the field as an established part of the social sciences (Fergnani and Chermack, 2021).

The Delphi projections selected for this study proved appropriate for the research focus and highly relevant, with an estimated average impact greater than five on a seven-point scale for all six projections (cf. Kluge et al., 2020). As the experts assessed the desirability of all projections as either somewhat high or high, it may be concluded that the addressed topics can be considered opportunities for the sports industry (cf. Schweizer et al., 2020). Looking at the probability assessments, it may be concluded that the posed projections represent realistic and conceivable future developments, given probability assessments between 60 and 88 % for all six projections. Despite these high expected probabilities, consensus was only reached for a few projections, which is an interesting finding as consensus projections typically fall within the 60 to 80 % expected probability range (Ogden et al., 2005; von der Gracht and Darkow, 2010). The experts reached consensus that consumption of sports content will continue to change significantly (e.g., shorter sequences or data-enriched content), which will impact the enormous (reciprocal) dependency of sports on television (cf. Miah et al., 2020; Rowe, 1996; Whannel, 2014) and continue the current trajectory (Chan-Olmsted and Xiao, 2019; Hutchins, 2012, 2018; Miah et al., 2020; Pirker, 2020). Experts also consent that at least ten times as much data will be collected in professional sports by 2030 compared to today, which is consistent with existing findings on the proliferation of data in sports in recent years (cf. Stein et al., 2017; Van Gisbergen et al., 2020). There was also consensus that new types of manager profiles in terms of backgrounds and skill sets in sports organizations would be desirable; a development similar to that in other industries (cf. Duclos-Bastías et al., 2021; González-Serrano et al., 2017; Govers and Amelsvoort, 2019; Kolbjørnsrud et al., 2016). With respect to impact, consensus was reached that new world records by 2030 in most Olympic disciplines due to technological advancements would have a high impact, which is consistent with Lippi et al. (2008) who argue that performance limits will increasingly depend on technological advances. The above-mentioned changes in the consumption of sports content would also have a high impact. As consensus on expected probability was only reached on two out of six projections, we performed a comprehensive dissent analysis (Beiderbeck et al., 2021b; Warth et al., 2013), which provided valuable insights. For example, we analyzed five different stakeholder groups and found that intra-group consensus was reached in almost two-thirds of all possible instances. At the same time, IQR scores for intra-group consensus projections were partially very low, indicating a very high level of consensus within groups in those cases.

To derive relevant insights from our results for academia and practice, we developed two scenarios (cf. Schmalz et al., 2021). Scenario one ("probable future") depicts a very high probability, high desirability, and high impact scenario. Probability assessments of all four projections in this scenario ranged from 79 to 88 % at an average of 83 %. This scenario pictures the future of all user groups in sports as largely technology-driven with significant changes in sports consumption, sports management, and athletes' use of data. Scenario two ("game changer") is a less probable, less desirable, and less impactful scenario compared to scenario one, but still has a relatively high expected probability (60–64 %) and impact and somewhat high desirability. It is in line with previous works that discuss a plateauing of records and performance (Balmer et al., 2012; Berthelot et al., 2008, 2010; Nevill et al., 2007) and the application of augmented, virtual, mixed, or extended reality experiences (Kim and Ko, 2019; Kunz and Santomier, 2020; Zollmann et al., 2019). While both scenarios describe developments in different areas of sports, they share common underlying themes. For all six projections, the experts acknowledged the impact of technology that is to be expected, which confirms the notion of previous works (cf. Frevel et al., 2020; Ratten, 2019; Schmidt, 2020). For the technological developments under discussion, the question is largely no longer how but when and to what extent.

In conclusion, all user groups in sports – athletes, consumers, and managers – may expect significant technology-driven changes by 2030. Different types of technologies can be differentiated in this development and not all of them will be relevant for all users. According to our experts, this technological progress can mostly be considered an opportunity rather than a threat and it can be a source of competitive advantage for both sports managers and athletes.

CRediT authorship contribution statement

Nicolas Frevel: Conceptualisation, Methodology, Software, Formal analysis, Investigation, Writing – Original Draft, Visualization, Project administration, Supervision. Prof. Dr. Sascha L. Schmidt: Conceptualisation, Methodology, Writing - Original Draft, Supervision. Daniel Beiderbeck: Conceptualisation, Methodology, Formal analysis, Data curation, Investigation.

Declaration of competing interest

The authors declare that there are no conflicts of interest regarding the publication of this paper.

Acknowledgements

We would like to thank Ben Shields, PhD, and Benjamin Penkert as well as three friendly reviewers for their supportive and constructive comments on previous versions of this manuscript. Furthermore, we sincerely thank our interviewees and participating experts for sharing their insights with us. Particularly in the midst of the COVID-19 crisis, this dedication is not to be taken for granted.

Appendix A

Table 4
Dissent analysis – desirability bias.

Projection	*r*	*df*	*t*	*p*
1: World records	0.25	90	2.49	0.014
2: Content formats	0.46	90	4.96	0.000
3: Manager profiles	0.43	90	4.55	0.000
4: Data collection	0.39	90	4.02	0.000
5: Data processing	0.58	90	6.70	0.000
6: Consumers/Fans	0.68	90	8.47	0.000

Note. Statistics describe correlation between expected probability and desirability of each respective projection.

Table 6
Stakeholder group analysis – results of the Mann-Whitney *U* test.

Group comparison	Projection	Dimension	*U*	*p*
Academia/Startup	1: World records	D	117.5	0.022
	2: Content formats	D	126	0.040
	5: Data processing	EP	133	0.063
	6: Consumers/Fans	EP	134	0.074
	6: Consumers/Fans	D	120	0.026
Academia/Consulting	4: Data collection	EP	248.5	0.078
Academia/Adjacencies	1: World records	D	123.5	0.090
Academia/Sports-centric	3: Manager profiles	D	201	0.072
	4: Data collection	EP	196	0.097
	6: Consumers/Fans	EP	100	0.095
Startup/Consulting	2: Content formats	D	256.5	0.057
	4: Data collection	EP	248	0.078
Startup/Adjacencies	2: Content formats	EP	241.5	0.065
	4: Data collection	D	243	0.059
	6: Consumers/Fans	EP	240	0.079
	6: Consumers/Fans	D	253	0.031
Startup/Sports-centric	1: World records	D	212.5	0.034
	4: Data collection	EP	200	0.067
	4: Data collection	EP	207.5	0.049
Consulting/Adjacencies	3: Manager profiles	D	116	0.077
Adjacencies/Sports-centric	1: World records	D	181	0.090
	3: Manager profiles	D	200.5	0.013

Note. EP = expected probability of occurrence; D = desirability of occurrence; I = impact in case of occurrence.

Table 12
Confirmatory factor analysis.

Construct	Mean	SD	CR	AVE
Attitude towards using technology (ATT)	5.86	1.03	0.91	0.71
Trust in technology (TRUST)	4.70	1.45	0.92	0.79

Note. SD = Standard deviation; CR = Composite Reliability; AVE = Average Variance Extracted; $CR = (\sum\lambda_i)^2 / ((\sum\lambda_i)^2 + \sum\theta_i)$; $AVE = \sum\lambda_i^2 / (\sum\lambda_i^2 + \sum\theta_i)$.

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